ROUTE B FOR THE MIDPOINT RVM SYSTEM: AN A PRIORI BOUND FOR QUASI-SOLUTIONS, AND A KADEC–AVDONIN CRITERION WITH BLOCKS OF VARIABLE LENGTH

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Abstract. We work with real lists $\boldsymbol{\kappa} = (\kappa_k)$ at bounded distance from the midpoint Riemann–von Mangoldt clock (τ_k) and with the rung system that couples such a list to atomic masses on $\{\log n\}$. (1) Route B: near-solutions of finite sections in a fixed box have a subsequence converging to an exact solution; with the von Mangoldt masses prescribed, the Laplace transform of the residual is identified as $\frac{1}{2}(P'_{\boldsymbol{\kappa}}/P_{\boldsymbol{\kappa}} - \xi'/\xi)$, every exact solution is the zeta list, and the resulting statement B′ is equivalent to the Riemann hypothesis; the equivalence is classical in substance. The existence part is given a certificate form for staged sections (Newton–Banach with a frozen right inverse), and Gonek's symmetrised Euler products give explicit lists in a tube around the clock. (2) For quasi-solutions with Laplace residual of size $\upsilon \log T$, the counting discrepancy with respect to the clock is $O((1+\upsilon)\log T)$ with absolute constants. (3) A Kadec–Avdonin criterion for the Muckenhoupt (A_2) property with averaging blocks of arbitrary, varying length; the $\frac{1}{4}$ theorems of Kadec and Avdonin are recovered. For the zeta list the remaining condition fails: unconditionally, no generating function of the unfolded zeta ordinates satisfies (A_2) on any horizontal line. (4) Computations: with the primes and the clock as the only input, exact square finite sections are certified up to height 487; the certified basin shrinks like T^{-2}.

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1. Introduction

The main manuscript constructs a synthetic zeta system on the midpoint clock τ_k, defined by $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, and reduces the remaining step to condition (D). Route B approaches (D) without constructing the zeta ordinates. One solves finite sections of the rung system (2.4), which imposes simultaneously a real critical spectrum and atomic arithmetic on $\{\log n\}$. If the residuals of these sections tend to zero in a fixed box, compactness yields an exact solution. The uniqueness theorem `thm:free-masses` then identifies that solution as the list of zeta ordinates with the von Mangoldt masses. In this scheme the ordinates γ_k are the output of an inverse problem, not an input. Section 3 makes this precise and isolates the two statements that remain open: existence of sections with vanishing residual, and uniqueness in the box without the third condition of $\mathscr{O}$. Section 3.1 shows that prescribing the von Mangoldt masses removes the second: the Laplace transform of the residual is identified (Lemma 3.10), and uniqueness follows from Hadamard's theorem and the functional equation, so that route B′ is equivalent to the Riemann hypothesis. Section 3.2 gives the existence part a certificate form: two computable quantities per stage, a right inverse of the Jacobian and the residual of the new tests, imply an exact solution of the stage. Section 3.3 shows that Gonek's symmetrised Euler products give explicit lists in a tube around the clock. Section 7 reports the computations that measure these quantities.

Section 4 proves an a priori bound for quasi-solutions. Section 5 proves a Kadec–Avdonin criterion with blocks of variable length, which is the tool needed for a Fredholm/stability formulation of the linearised problem. Section 6 applies that criterion to lists near the clock and proves that, for the zeta list, the uniform version fails.

References of the form `thm:free-masses` point to labels of the main manuscript `rvm_zeta_pi_mertens_v19`, of which this note is also a section; the notation is that of the main manuscript. Except in Section 6.3, where the zeta list is the object of study, nothing below uses the ordinates of ζ.

2. Setting

Throughout, $q(s) := s(s-1)$ and $G(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)$. The smooth clock count is

$$\mathcal{N}_{\mathrm{RVM}}(t) := \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}, \qquad \mathcal{N}'_{\mathrm{RVM}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} \qquad (t>0). \tag{2.1}$$

It is strictly increasing on $[2\pi,\infty)$, with $\mathcal{N}_{\mathrm{RVM}}(2\pi) = -\frac{1}{8}$, and the clock points are the solutions $\tau_k > 2\pi$ of $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac{1}{2}$, $k \ge 1$ ($\tau_1 = 14.5213\ldots$). We write $n_{\mathrm{RVM}}(t) := \#\{k \ge 1 : \tau_k \le t\}$. Since $\mathcal{N}_{\mathrm{RVM}}$ is increasing,

$$n_{\mathrm{RVM}}(t) = \Big\lfloor \mathcal{N}_{\mathrm{RVM}}(t) + \tfrac{1}{2} \Big\rfloor, \qquad |n_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(t)| \le \tfrac{1}{2} \qquad (t \ge 2\pi). \tag{2.2}$$

Definition 2.1 (Admissible lists)**.** For $C > 0$, a list $\boldsymbol{\kappa} = (\kappa_k)_{k\ge1}$ is *C-admissible* if it is real and nondecreasing, $\kappa_1 \ge 2\pi$, and $|\kappa_k - \tau_k| \le C$ for all k. Put $N_{\boldsymbol{\kappa}}(t) := \#\{k : \kappa_k \le t\}$ (with multiplicity) and

$$P_{\boldsymbol{\kappa}}(s) := \prod_{k\ge1}\Big(1 + \frac{q(s)}{\frac{1}{4}+\kappa_k^2}\Big), \qquad \zeta_{\boldsymbol{\kappa}} := \frac{P_{\boldsymbol{\kappa}}}{G}. \tag{2.3}$$

The clock list $\boldsymbol{\tau} = (\tau_k)$ gives $\zeta_{\boldsymbol{\tau}} = \zeta_{\mathrm{RVM}}$.

$P_{\boldsymbol{\kappa}}$ is entire and real on $\mathbb{R}$, and $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$. Its zeros are exactly $\frac{1}{2} \pm i\kappa_k$, with multiplicity, and it is positive on $[-1, 2]$, because $q([-1, 2]) = [-\frac{1}{4}, 2]$ and $\kappa_k^2 \geq 4\pi^2$. Writing $s = \frac{1}{2} + w$, one has $\frac{1}{4} + \kappa^2 + q(s) = \kappa^2 + w^2$.

The *rung system* `eq:rung-system` for a list $\boldsymbol{\kappa}$ and real masses $\mathsf{a} = (\mathsf{a}_n)_{n\geq 2}$ is

$$\sum_{k\geq 1} \cos(\kappa_k u) + \sum_{n\geq 2} \mathsf{a}_n\, \delta(u - \log n) = \mathsf{k}(u) \qquad \text{in } \mathcal{D}'(0,\infty), \tag{2.4}$$

with the fixed distribution k of `eq:transport-defs`. Its residual is the distribution $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a})$ defined, for $\varphi \in C_c^\infty(0,\infty)$, by

$$\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle := \sum_{k\geq 1} \mathsf{c}_\varphi(\kappa_k) + \sum_{n\geq 2} \mathsf{a}_n \varphi(\log n) - \langle \mathsf{k}, \varphi\rangle, \qquad \mathsf{c}_\varphi(t) := \int_0^\infty \varphi(u)\cos(tu)\, du. \tag{2.5}$$

Two integrations by parts give $|\mathsf{c}_\varphi(t)| \leq \|\varphi''\|_{L^1}\, t^{-2}$. Since $\kappa_k \geq \tau_k - C$ and $\tau_k \gg k/\log(k+1)$, the series in (2.5) converges absolutely. The mass sum is finite, because $\operatorname{supp}\varphi \subset [u_1, u^*]$ with $0 < u_1 < u^*$ leaves only $n \leq e^{u^*}$.

3. Route B: compactness, selection and the canonical box

For $C, \mathbb{M} > 0$ let $\mathbb{K}(C, \mathbb{M})$ be the set of pairs $(\boldsymbol{\kappa}, \mathsf{a})$ with $\boldsymbol{\kappa}$ C-admissible and $|\mathsf{a}_n| \leq \mathbb{M}$ for all n. A *finite section* is an element of $\mathbb{K}(C, \mathbb{M})$ with $\kappa_k = \tau_k$ for $k > k^*$ and $\mathsf{a}_n = 0$ for $n > n^*$, for some k^*, n^*.

Fix finite sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots \subset C_c^\infty(0,\infty)$ whose union $\mathscr{T}_\infty$ is dense in $C_c^\infty(0,\infty)$. Such sets exist because $C_c^\infty(0,\infty)$ is separable for its usual LF topology: it is the countable strict inductive limit of the separable Fréchet spaces $C^\infty_{[1/j,j]}$, and a countable union of countable dense subsets of these is dense. For $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$ put

$$\mathsf{s}_m(\boldsymbol{\kappa}, \mathsf{a}) := \max_{\varphi \in \mathscr{T}_m} |\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle|. \tag{3.1}$$

Definition 3.1 (Statement B)**.** *Statement* $B(C, \mathbb{M})$ says that there are finite sections $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C, \mathbb{M})$ with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$.

Lemma 3.2 (Canonical box of the exact solution)**.** *Let* $\mathbb{M}_* := \max_{n\geq 2} \Lambda(n)/(2\sqrt{n})$. *Then* $\mathbb{M}_* = \log 7/(2\sqrt{7}) = 0.367742\ldots < 1/e$. *If condition (D) holds, the pair* $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, *with* $\mathsf{a}^\zeta_n := \Lambda(n)/(2\sqrt{n})$, *solves* (2.4) *and lies in* $\mathbb{K}(C_\eta, \mathbb{M}_*)$, *where* C_η *is the constant of* `lem:bounded-displacement`.

Proof. $x \mapsto \log x/(2\sqrt{x})$ increases on $[1, e^2]$ and decreases on $[e^2, \infty)$, with $e^2 = 7.389\ldots$. Hence among primes its maximum is at $p = 7$ or $p = 11$; the values are $0.367742\ldots$ and $0.361540\ldots$. A prime power p^j with $j \geq 2$ gives $\log p/(2p^{j/2}) < \log p/(2\sqrt{p}) \leq \mathbb{M}_*$. The bound by $1/e$ is the continuous maximum, attained at $x = e^2$. Under (D), $\zeta_{Z1} = \zeta$, so all zeros of ζ are $\frac{1}{2} \pm i\gamma_k$. The rung system with list $\boldsymbol{\gamma}$ and masses a^ζ is then the distributional explicit formula `thm:distributional-EF`. Finally, $|\gamma_k - \tau_k| \leq C_\eta$ by `lem:bounded-displacement`, and $\gamma_1 = 14.13\ldots \geq 2\pi$. □

Corollary 3.3 ((D) implies B)**.** *If (D) holds, then* $B(C, \mathbb{M})$ *holds for every* $C \geq C_\eta$ *and* $\mathbb{M} \geq \mathbb{M}_*$.

Proof. Let $(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)})$ be the truncation of $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$. Its list is the nondecreasing rearrangement of the multiset $\{\gamma_1, \ldots, \gamma_K\} \cup \{\tau_{K+1}, \tau_{K+2}, \ldots\}$, and its masses are a^ζ_n for $n \leq n^*$ and 0 beyond. By Lemma 3.17 (Section 3.2) the rearranged list satisfies $|\kappa_k - \tau_k| \leq C_\eta$, coincides with (τ_k) beyond some index, and $\mathfrak{R}$ depends only on the multiset. So this is a finite section in $\mathbb{K}(C_\eta, \mathbb{M}_*)$. For fixed φ, $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(k^*,n^*)}, \mathsf{a}^{(k^*,n^*)}), \varphi\rangle - \langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \varphi\rangle = \sum_{k>k^*} (\mathsf{c}_\varphi(\tau_k) - \mathsf{c}_\varphi(\gamma_k))$ as soon as $e^{u^*} \leq n^*$. This tends to 0 as $k^* \to \infty$. Since $\mathscr{T}_m$ is finite and $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ by Lemma 3.2, k^*_m, N_m can be chosen with $\mathsf{s}_m \leq 1/m$. □

Proposition 3.4 (Compactness and passage to the limit)**.** *Let* $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \in \mathbb{K}(C, \mathbb{M})$ *with* $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}) \to 0$*. Then a subsequence converges coordinatewise to some* $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$*, and every such limit satisfies* $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ *in* $\mathcal{D}'(0, \infty)$*.*

Proof. Each $\kappa_k^{(m)}$ lies in the compact interval $[\tau_k - C, \tau_k + C]$ and each $\mathsf{a}_n^{(m)}$ in $[-\mathbb{M}, \mathbb{M}]$. A diagonal extraction over the countably many coordinates gives a coordinatewise convergent subsequence. The limit is nondecreasing, satisfies $\kappa_1 \geq 2\pi$, $|\kappa_k - \tau_k| \leq C$ and $|\mathsf{a}_n| \leq \mathbb{M}$, and therefore lies in $\mathbb{K}(C, \mathbb{M})$.

Continuity of each pairing. Fix φ with support in $[u_1, u^*]$. The mass part of (2.5) involves only $n \leq e^{u^*}$ and converges. For the spectral part, $|\mathsf{c}_\varphi(\kappa_k^{(m)})| \leq \|\varphi''\|_1(\tau_k - C)^{-2}$ when $\tau_k > 2C$, and $|\mathsf{c}_\varphi| \leq \|\varphi\|_1$ for the finitely many other k. The majorant is summable, so dominated convergence for series gives $\sum_k \mathsf{c}_\varphi(\kappa_k^{(m)}) \to \sum_k \mathsf{c}_\varphi(\kappa_k)$. Hence $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle \to \langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle$ along the subsequence.

Vanishing. If $\varphi \in \mathscr{T}_\infty$, then $\varphi \in \mathscr{T}_m$ for all large m, so $|\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)}), \varphi\rangle| \leq \mathsf{s}_m \to 0$ and $\langle \mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}), \varphi\rangle = 0$. The limit $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a})$ is a single distribution, hence continuous on $C_c^\infty(0, \infty)$. It vanishes on the dense set $\mathscr{T}_\infty$, so it vanishes identically. No uniform bound on the residuals is needed for this step. □

Hypothesis 3.5 (Uniqueness in the box)**.** $U(C, \mathbb{M})$: the only $(\boldsymbol{\kappa}, \mathsf{a}) \in \mathbb{K}(C, \mathbb{M})$ with $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}) = 0$ is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$.

Remark 3.6 (What is proved and what is open in U). `thm:free-masses` proves uniqueness for lists in $\mathscr{O}$. Besides the tube and the ordering, $\mathscr{O}$ (`eq:list-space`) carries a third condition, $|\int (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}})| \leq C \log \mathrm{X}$, and a coordinatewise limit of sections does not inherit it. Hypothesis 3.5 is therefore the statement that `thm:free-masses` holds for exact solutions in $\mathbb{K}(C, \mathbb{M})$ without the third condition. That condition is used to bound $\zeta_{\boldsymbol{\kappa}}$ on half-planes. For an exact solution, Lemma 4.3 below with $\widehat{\mathsf{R}} \equiv 0$ gives that bound directly on $\Re s \geq 2$. Whether this suffices for the rest of the proof of `thm:free-masses` has to be checked line by line against that proof. This is not done here, and Hypothesis 3.5 is not claimed.

Corollary 3.7 (Selection of the zeta list)**.** *Assume* $B(C, \mathbb{M})$ *and* $U(C, \mathbb{M})$*, and let* $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^{(m)})$ *be any sequence in* $\mathbb{K}(C, \mathbb{M})$ *with* $\mathsf{s}_m \to 0$*. Then the whole sequence converges coordinatewise to* $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$*. In particular* $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$*, that is, (D) holds.*

Proof. By Proposition 3.4, every subsequence has a further subsequence converging to a zero of $\mathfrak{R}$ in $\mathbb{K}(C, \mathbb{M})$. By $\mathrm{U}(C, \mathbb{M})$ that zero is $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$, so the whole sequence converges to it. Finally we show that $\mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta) = 0$ implies (D). By Lemma 3.10 (Section 3.1), $\widehat{\mathsf{R}}$ of (4.1) is the Laplace transform of a distribution of finite order on $[0, \infty)$ whose restriction to $(0, \infty)$ is $\mathfrak{R}$. If $\mathfrak{R} = 0$, that distribution is supported at $\{0\}$, so $\widehat{\mathsf{R}}$ is a polynomial $p(s)$. Since $\sum_n \mathsf{a}_n^\zeta n^{\frac{1}{2}-s} = -\frac{1}{2}\zeta'/\zeta(s)$, this reads $P_\gamma'/P_\gamma - (G\zeta)'/(G\zeta) = 2p$, so $P_\gamma = c\, G\zeta\, e^Q$ with $Q' = 2p$. Here $G\zeta = 2\xi$ and P_γ are entire of order 1, so $e^Q = P_\gamma/(2c\,\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \leq 1$. Both P_γ and ξ are invariant under $s \mapsto 1 - s$, so $e^{Q(1-s)} = e^{Q(s)}$, which forces Q to be constant. Hence $\zeta_\gamma = c'\zeta$. Both functions have residue 1 at $s = 1$ ($P_\gamma(1) = 1$ and $G(s) \sim s - 1$), so $c' = 1$. This is (D), because $\zeta_\gamma = \zeta_{Z1}$. □

Remark 3.8 (Logical status of route B). Corollaries 3.3 and 3.7 give (D) $\Rightarrow$ $\mathrm{B}(C_\eta, \mathbb{M}_*)$ and [$\mathrm{B}(C, \mathbb{M})$ and $\mathrm{U}(C, \mathbb{M})$] $\Rightarrow$ (D). Route B therefore consists of two independent pieces. The first is existence, B, which is not weaker than (D). The second is uniqueness without the third condition of $\mathscr{O}$, Hypothesis 3.5, which does not involve ζ. Compactness itself is not an issue in the box: Proposition 3.4 uses only the imposed bounds. In this scheme the ordinates γ_k never enter the construction; they appear only as the unique admissible zero of $\mathfrak{R}$. Lemma 3.2 does *not* show that near-solutions of sections lie in $\mathbb{K}(C_\eta, \mathbb{M}_*)$: it locates only the exact solution. Restricting the sections to that box is a choice, justified a posteriori by Corollary 3.3.

3.1. **Route B with prescribed von Mangoldt masses.** The masses $\mathsf{a}_n^{\zeta} = \Lambda(n)/(2\sqrt{n})$ are known and do not depend on (D) or on the ordinates of ζ. Prescribing them leaves the ordinates as the only unknowns, and the uniqueness question of Hypothesis 3.5 disappears. The tool is the identification of the Laplace transform of the residual. In the previous version this was a convention; here it is a lemma.

Laplace transforms of distributions. Let $\mathscr{T} \in \mathcal{D}'(\mathbb{R})$ have support in $[0,\infty)$ and suppose that $e^{-au}\mathscr{T}$ is a tempered distribution for some $a \in \mathbb{R}$. For $\Re s > a + \frac{1}{2}$ put $z := s - \frac{1}{2}$ and

$$\mathcal{L}\mathscr{T}(s) := \big\langle \mathscr{T},\ \mathsf{j}(u)\, e^{-zu} \big\rangle, \tag{3.2}$$

where $\mathsf{j} \in C^\infty(\mathbb{R})$ equals 1 on $[-1,\infty)$ and 0 on $(-\infty,-2]$; the value does not depend on j, and $\mathcal{L}\mathscr{T}$ is holomorphic there [20, Ch. 8]. The shift by $\frac{1}{2}$ in z is the normalisation of (4.1). Since j' vanishes on the support of $\mathscr{T}$, $\mathcal{L}(\partial_u \mathscr{T})(s) = z\,\mathcal{L}\mathscr{T}(s)$; also $\mathcal{L}\delta_0 = 1$, $\mathcal{L}\delta_0' = z$, and for a locally integrable g on $[0,\infty)$ with $|g(u)| \le ce^{au}$, $\mathcal{L}(g\mathbf{1}_+)(s) = \int_0^\infty e^{-zu} g(u)\,du$. We write $\mathbf{1}_+ := \mathbf{1}_{[0,\infty)}$ and $\operatorname{fp}\frac{\mathbf{1}_+}{u}$ for the distribution $\varphi \mapsto \int_0^1 \frac{\varphi(u)-\varphi(0)}{u}\,du + \int_1^\infty \frac{\varphi(u)}{u}\,du$, of order 1, supported in $[0,\infty)$, with restriction $1/u$ to $(0,\infty)$.

Lemma 3.9 (The Archimedean distribution)**.** *Let* $\mathfrak{g}(u) := \frac{e^{-u/2}}{e^{2u}-1} - \frac{1}{2u}$ *for* $u > 0$*;* $\mathfrak{g}$ *is bounded,* $\mathfrak{g}(0^+) = -\frac{3}{4}$*, and* $\mathfrak{g}(u) = O(e^{-u/2})$*. Put* $\mathrm{c}_{\mathscr{K}} := -\frac{1}{4}\log(2\pi e^{\gamma_E})$*,* γ_E *Euler's constant, and*

$$\mathscr{K} := \tfrac{1}{2}e^{u/2}\mathbf{1}_+ - \tfrac{1}{2}\,\mathfrak{g}\,\mathbf{1}_+ - \tfrac{1}{4}\operatorname{fp}\frac{\mathbf{1}_+}{u} + \mathrm{c}_{\mathscr{K}}\,\delta_0. \tag{3.3}$$

Then $\mathscr{K}$ *is a distribution of order* 1 *supported in* $[0,\infty)$*, its restriction to* $(0,\infty)$ *is the function* k *of* `eq: transport-defs`*, and*

$$\mathcal{L}\mathscr{K}(s) = \tfrac{1}{2}\,\frac{G'}{G}(s) \qquad (\Re s > 1). \tag{3.4}$$

Proof. On $(0,\infty)$, $\frac{1}{2}e^{u/2} - \frac{1}{2}\mathfrak{g}(u) - \frac{1}{4u} = \frac{1}{2}\big(e^{u/2} - \frac{e^{-u/2}}{e^{2u}-1}\big) = \mathsf{k}(u)$. The expansion $\frac{e^{-u/2}}{e^{2u}-1} = \frac{1}{2u} - \frac{3}{4} + O(u)$ gives the behaviour of $\mathfrak{g}$ at 0. With $z = s - \frac{1}{2}$ and $\Re s > 1$: $\mathcal{L}(\frac{1}{2}e^{u/2}\mathbf{1}_+) = \frac{1}{2(z-\frac{1}{2})} = \frac{1}{2(s-1)}$. Gauss's integral $\frac{\Gamma'}{\Gamma}(x) = \int_0^\infty \big(\frac{e^{-t}}{t} - \frac{e^{-xt}}{1-e^{-t}}\big)dt$ $(\Re x > 0)$ for the logarithmic derivative of Γ [19, Ch. XII], with $t = 2u$ and $x = \frac{s}{2}$, reads

$$\tfrac{1}{2}\frac{\Gamma'}{\Gamma}\big(\tfrac{s}{2}\big) = \int_0^\infty \Big(\frac{e^{-2u}}{2u} - \frac{e^{-su}}{1-e^{-2u}}\Big)du = \int_0^\infty \Big(\frac{e^{-2u}}{2u} - e^{-zu}\frac{e^{-u/2}}{e^{2u}-1}\Big)du - \frac{1}{s}, \tag{3.5}$$

because $\frac{e^{-su}}{1-e^{-2u}} = e^{-zu}\big(\frac{e^{-u/2}}{e^{2u}-1} + e^{-u/2}\big)$ and $\int_0^\infty e^{-zu}e^{-u/2}du = \frac{1}{s}$. Writing $\frac{e^{-u/2}}{e^{2u}-1} = \mathfrak{g}(u) + \frac{1}{2u}$ and using Frullani's integral $\int_0^\infty \frac{e^{-2u}-e^{-zu}}{u}\,du = \log\frac{z}{2}$ (both sides vanish at $z=2$ and have derivative $\int_0^\infty e^{-zu}du = 1/z$ in z, for $\Re z > 0$), the integral in (3.5) equals $-\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) + \frac{1}{2}\log\frac{z}{2}$. Hence

$$\tfrac{1}{2}\frac{G'}{G}(s) = \frac{1}{2s} + \frac{1}{2(s-1)} - \frac{1}{4}\log\pi + \frac{1}{4}\frac{\Gamma'}{\Gamma}\big(\tfrac{s}{2}\big) = \mathcal{L}(\tfrac{1}{2}e^{u/2}\mathbf{1}_+)(s) - \tfrac{1}{2}\mathcal{L}(\mathfrak{g}\mathbf{1}_+)(s) + \tfrac{1}{4}\log\frac{z}{2} - \tfrac{1}{4}\log\pi. \tag{3.6}$$

Finally $\mathcal{L}(\operatorname{fp}\frac{\mathbf{1}_+}{u})(s) = \int_0^1 \frac{e^{-zu}-1}{u}\,du + \int_1^\infty \frac{e^{-zu}}{u}\,du = -\gamma_E - \log z$: the left side has derivative $-\int_0^\infty e^{-zu}du = -1/z$ in z, and its value at $z=1$ is $\int_0^1 \frac{e^{-u}-1}{u}\,du + \int_1^\infty \frac{e^{-u}}{u}\,du = -\gamma_E$, the classical integral representation of Euler's constant [19, Ch. XII]. Substituting, $\frac{1}{4}\log\frac{z}{2} = -\frac{1}{4}\mathcal{L}(\operatorname{fp}\frac{\mathbf{1}_+}{u}) - \frac{1}{4}(\gamma_E + \log 2)$, and (3.6) becomes (3.4) with the constant $\mathrm{c}_{\mathscr{K}}$. (The identity was also checked numerically to 25 digits at four points s; file `laplace_check.py`.) □

Lemma 3.10 (Laplace transform of the residual)**.** *Let* $\boldsymbol{\kappa}$ *be C-admissible and put* $\mathfrak{H}_{\boldsymbol{\kappa}}(u) := \sum_k \kappa_k^{-2}\cos(\kappa_k u)$*, which converges absolutely and uniformly on* $\mathbb{R}$ *since* $\mathfrak{H}_{\boldsymbol{\kappa}}(0) = \sum_k \kappa_k^{-2} < \infty$*,*

and

$$\mathscr{S}_{\boldsymbol{\kappa}} := -\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+) + \mathfrak{H}_{\boldsymbol{\kappa}}(0)\,\delta_0', \qquad \mathscr{A}_\zeta := \sum_{n\geq 2} \mathsf{a}_n^\zeta\,\delta_{\log n}, \qquad \mathscr{E}_{\boldsymbol{\kappa}} := \mathscr{S}_{\boldsymbol{\kappa}} + \mathscr{A}_\zeta - \mathscr{K}. \tag{3.7}$$

Then $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of order at most 2, supported in $[0,\infty)$, whose restriction to $(0,\infty)$ is $\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}^\zeta)$, and for $\Re s > 1$

$$\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}(s) = \widehat{\mathsf{R}}(s) = \tfrac{1}{2}\Big(\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}} - \frac{\xi'}{\xi}\Big)(s), \qquad \xi := \tfrac{1}{2}G\zeta, \tag{3.8}$$

where $\widehat{\mathsf{R}}$ is (4.1) *with the masses a^ζ for all n. The same holds for a finite section $(\boldsymbol{\kappa},\mathsf{a})$ with $\mathscr{A}_\zeta$ replaced by $\sum_{n\leq n^*}\mathsf{a}_n\delta_{\log n}$ and $-\frac{1}{2}\xi'/\xi$ by $-\frac{1}{2}G'/G + \sum_{n\leq n^*}\mathsf{a}_n n^{\frac{1}{2}-s}$.*

Proof. Spectral part. $\kappa_k \geq \tau_k - C \gg k/\log(k+1)$, so $\mathfrak{H}_{\boldsymbol{\kappa}}(0) < \infty$ and $\mathfrak{H}_{\boldsymbol{\kappa}}$ is continuous and bounded by $\mathfrak{H}_{\boldsymbol{\kappa}}(0)$. In $\mathcal{D}'(0,\infty)$ a uniformly convergent series may be differentiated termwise, so $-\mathfrak{H}_{\boldsymbol{\kappa}}'' = \sum_k \cos(\kappa_k u)$ there; the term $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ is supported at 0. Hence $\mathscr{S}_{\boldsymbol{\kappa}}|_{(0,\infty)} = \sum_k \cos(\kappa_k u)$, and $\mathscr{S}_{\boldsymbol{\kappa}}$ has order at most 2. For $\Re z > 0$, Fubini (the double integral $\sum_k \kappa_k^{-2}\int_0^\infty e^{-\Re z\, u}du$ is finite) gives $\mathcal{L}(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)(s) = \sum_k \kappa_k^{-2}\frac{z}{z^2+\kappa_k^2}$, so

$$\mathcal{L}\mathscr{S}_{\boldsymbol{\kappa}}(s) = -z^2\sum_k \frac{z}{\kappa_k^2(z^2+\kappa_k^2)} + z\sum_k \frac{1}{\kappa_k^2} = \sum_k \frac{z}{z^2+\kappa_k^2} = \tfrac{1}{2}\frac{P_{\boldsymbol{\kappa}}'}{P_{\boldsymbol{\kappa}}}(s), \tag{3.9}$$

because $\frac{z}{\kappa^2} - \frac{z^3}{\kappa^2(z^2+\kappa^2)} = \frac{z}{z^2+\kappa^2}$, and $P_{\boldsymbol{\kappa}} = \prod_k \frac{z^2+\kappa_k^2}{\frac{1}{4}+\kappa_k^2}$ converges locally uniformly, so its logarithmic derivative is the termwise sum $\sum_k \frac{2z}{z^2+\kappa_k^2}$.

Arithmetic part. On $[0,u^*]$ only $n \leq e^{u^*}$ contribute, so $\mathscr{A}_\zeta$ is a locally finite measure, of order 0, and $e^{-au}\mathscr{A}_\zeta$ is a finite measure for $a > \frac{1}{2}$. For $\Re s > 1$, $\mathcal{L}\mathscr{A}_\zeta(s) = \sum_n \frac{\Lambda(n)}{2\sqrt{n}}n^{-z} = \frac{1}{2}\sum_n \Lambda(n)n^{-s} = -\frac{1}{2}\frac{\zeta'}{\zeta}(s)$.

Assembly. By Lemma 3.9, $\mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}} = \frac{1}{2}P_{\boldsymbol{\kappa}}'/P_{\boldsymbol{\kappa}} - \frac{1}{2}\zeta'/\zeta - \frac{1}{2}G'/G$, and $(G\zeta)'/(G\zeta) = \xi'/\xi$ since $\xi = \frac{1}{2}G\zeta$. The restriction to $(0,\infty)$ is $\sum_k \cos(\kappa_k u) + \sum_n \mathsf{a}_n^\zeta\delta(u-\log n) - \mathsf{k}(u)$, which is (2.5). The finite-section case is identical with a finite sum. □

Remark 3.11 (The boundary term). Without the correction $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$ in (3.7), the transform of $-\partial_u^2(\mathfrak{H}_{\boldsymbol{\kappa}}\mathbf{1}_+)$ is $\frac{1}{2}P_{\boldsymbol{\kappa}}'/P_{\boldsymbol{\kappa}} - (s-\frac{1}{2})\mathfrak{H}_{\boldsymbol{\kappa}}(0)$: the second derivative of the cut-off function produces $\mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0' = \mathfrak{H}_{\boldsymbol{\kappa}}(0)\delta_0'$. Either normalisation gives the same restriction to $(0,\infty)$ and differs by a polynomial in s, which is harmless in the argument below. Both $\mathscr{K}$ and $\mathscr{S}_{\boldsymbol{\kappa}}$ carry singular terms at $u=0$; this is why the rung system is posed in $\mathcal{D}'(0,\infty)$ and not on $[0,\infty)$.

Proposition 3.12 (Uniqueness with prescribed masses)**.** *Let $\boldsymbol{\kappa}$ be C-admissible for some C, and suppose $\mathfrak{R}(\boldsymbol{\kappa},\mathsf{a}^\zeta) = 0$ in $\mathcal{D}'(0,\infty)$. Then all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, and $\boldsymbol{\kappa} = \boldsymbol{\gamma}$.*

Proof. By Lemma 3.10, $\mathscr{E}_{\boldsymbol{\kappa}}$ is a distribution of finite order supported in $[0,\infty)$ whose restriction to $(0,\infty)$ vanishes, so it is supported at $\{0\}$. A distribution of finite order supported at a point is a finite linear combination of δ_0 and its derivatives [10, Thm. 2.3.4], so $\widehat{\mathsf{R}} = \mathcal{L}\mathscr{E}_{\boldsymbol{\kappa}}$ is a polynomial p in $z = s - \frac{1}{2}$. By (3.8), $(P_{\boldsymbol{\kappa}}e^{-Q}/\xi)' = 0$ on $\Re s > 1$ with $Q' = 2p$, Q a polynomial. Hence $P_{\boldsymbol{\kappa}} = c\,\xi\,e^Q$ there with $c \neq 0$, and by analytic continuation on all of $\mathbb{C}$. In the variable $w = s - \frac{1}{2}$, $P_{\boldsymbol{\kappa}}$ is a constant multiple of $\prod_k(1 + w^2/\kappa_k^2)$, and $N_{\boldsymbol{\kappa}}(t) \ll t\log t$, so $P_{\boldsymbol{\kappa}}$ has order 1; ξ has order 1. Thus $e^Q = P_{\boldsymbol{\kappa}}/(c\xi)$ is entire, zero-free and of order at most 1, and $\deg Q \leq 1$ by Hadamard's theorem. Since $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$ and $\xi(1-s) = \xi(s)$, $e^{Q(1-s)} = e^{Q(s)}$; for $Q(s) = a + bs$ this forces $b(1-2s) \in 2\pi i\mathbb{Z}$ for all s, so $b = 0$. Therefore $P_{\boldsymbol{\kappa}} = c'\xi$. The zeros of $P_{\boldsymbol{\kappa}}$ are $\frac{1}{2} \pm i\kappa_k$, so every zero of ξ, that is, every nontrivial zero of ζ, has real part $\frac{1}{2}$, and the ordinates counted with multiplicity are the κ_k. Both lists are nondecreasing, so $\boldsymbol{\kappa} = \boldsymbol{\gamma}$. □

Definition 3.13 (Statement B′)**.** *Statement $B'(C)$* says that there are C-admissible lists $\boldsymbol{\kappa}^{(m)}$, each equal to the clock beyond some finite index, with $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) \to 0$.

Corollary 3.14 (Route B′)**.** (a) *If $B'(C)$ holds for some C, then every sequence as in Definition 3.13 converges coordinatewise to $\boldsymbol{\gamma}$, and all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$.*
(b) *If all nontrivial zeros of ζ lie on $\Re s = \frac{1}{2}$, then $B'(C)$ holds for every $C \geq C_\eta$.*
Hence the Riemann hypothesis is equivalent to $B'(C)$ for some $C < \infty$.

Proof. (a) $|\mathsf{a}_n^\zeta| \leq \mathbb{M}_*$ by the elementary computation in Lemma 3.2, so $(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta) \in \mathbb{K}(C, \mathbb{M}_*)$. By Proposition 3.4, every subsequence has a further subsequence converging coordinatewise to some $(\boldsymbol{\kappa}, \mathsf{a}^\zeta)$ with $\mathfrak{R}(\boldsymbol{\kappa}, \mathsf{a}^\zeta) = 0$ and $\boldsymbol{\kappa}$ C-admissible; the masses do not move. By Proposition 3.12, $\boldsymbol{\kappa} = \boldsymbol{\gamma}$ and the zeros lie on the line. Since every subsequence has a further subsequence with the same limit, the whole sequence converges. (b) Under this hypothesis the rung system with $(\boldsymbol{\gamma}, \mathsf{a}^\zeta)$ is the distributional explicit formula `thm:distributional-EF`, and $|\gamma_k - \tau_k| \leq C_\eta$ by `lem:bounded-displacement`. The truncations in the proof of Corollary 3.3, with the masses kept equal to a^ζ, give the sections. □

Remark 3.15 (What prescribing the masses changes, and what it does not). Hypothesis 3.5 is no longer needed: uniqueness holds for every list at bounded distance from the clock, not only in $\mathscr{O}$, and it follows from Hadamard's theorem and the functional equation instead of `thm:free-masses`. Route B′ has a single open piece, the existence statement B′, which by Corollary 3.14 is equivalent to the Riemann hypothesis. With prescribed masses the whole window $(0, u^*)$ constrains the ordinates; the part above $\log \Omega$ can no longer be absorbed by the masses, as it could in Heuristic 4.12.

The equivalence itself is classical in substance. The prime side of the rung system is the prime side of the explicit formula of Riemann and von Mangoldt in the form of Guinand and Weil [8, 18], and the uniqueness is Hadamard's factorisation. Arias de Reyna [1] proves that the Riemann hypothesis is equivalent to the temperedness of the measure $-\sum_n \frac{\Lambda(n)}{\sqrt{n}}(\delta_{\log n} + \delta_{-\log n}) + 2\cosh(x/2)\,dx$, the same prime side up to the factor 2 and the symmetrisation, and identifies its Fourier transform with $\sum_\gamma \delta_{\gamma/2\pi} - 2\vartheta'(2\pi t)\,dt$ under that hypothesis; by partial summation his criterion is von Koch's $\sum_{n \leq x} \Lambda(n) = x + O(x^{1/2}\log^2 x)$. Statement B′ is the finite-section form of the same equivalence: it asks for a real spectrum in a tube around the clock that reproduces the prime side against a dense family of tests. It is not a weaker intermediate statement, and the finite sections are not more constrained than the global statement; what they offer is a formulation in which each stage is a finite-dimensional problem with computable certificates (Section 3.2) and explicit candidates (Section 3.3). Dyson's suggestion that the zeros form a one-dimensional quasicrystal, and the classification of crystalline measures by Kurasov and Sarnak [12] and Olevskii and Ulanovskii [15], concern the same object; the zeta list is not in those classes because of its density, as Arias de Reyna points out.

3.2. **Staged construction: a certificate theorem and a graded extension.**

Definition 3.16 (Staged sections)**.** Fix $u_1^* < u_2^* < \cdots \to \infty$, finite test sets $\mathscr{T}_1 \subset \mathscr{T}_2 \subset \cdots$ supported in $(0, u_m^*)$ with dense union as in Section 3, $\mathrm{t}_m := |\mathscr{T}_m|$, and $k_m^* \to \infty$. Stage m starts from a list $\boldsymbol{\kappa}^{(m,0)}$ that equals the clock beyond some finite index, moves the coordinates $k \leq k_m^*$ by $x \in \mathbb{R}^{k_m^*}$, and has masses a^ζ. Its residual map is

$$\mathbb{F}_m : \mathbb{R}^{k_m^*} \to \mathbb{R}^{\mathrm{t}_m}, \qquad \mathbb{F}_m(x)_\varphi := \big\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m,0)} + x, \mathsf{a}^\zeta), \varphi \big\rangle \quad (\varphi \in \mathscr{T}_m), \tag{3.10}$$

where $(\boldsymbol{\kappa} + x)_k = \kappa_k + x_k$ for $k \leq k_m^*$ and κ_k otherwise, and its Jacobian at 0 is

$$(\mathbb{J}_m)_{\varphi,k} = \partial_{x_k} \mathbb{F}_m(0)_\varphi = -\int_0^\infty \varphi(u)\, u \sin(\kappa_k^{(m,0)} u)\, du \qquad (\varphi \in \mathscr{T}_m,\ k \leq k_m^*). \tag{3.11}$$

We use the norms $\|x\|_\infty$ on $\mathbb{R}^{k_m^*}$ and $\mathbb{R}^{\mathrm{t}_m}$ and the induced operator norms $\|\cdot\|_{\infty\to\infty}$ (maximal absolute row sum). Put

$$\mathsf{h}_m := \|\mathbb{F}_m(0)\|_\infty, \qquad \mathbb{H}_m := k_m^* \max_{\varphi\in\mathscr{T}_m} \int_0^\infty |\varphi(u)|\, u^2\, du. \tag{3.12}$$

Lemma 3.17 (Monotone matching)**.** *Let $(\mathrm{a}_k)_{k\ge1}$ be real with $|\mathrm{a}_k-\tau_k|\le C$ for all k, and let $(\mathrm{a}_{(k)})$ be its nondecreasing rearrangement. Then $|\mathrm{a}_{(k)}-\tau_k|\le C$ for all k. If moreover $\mathrm{a}_k=\tau_k$ for $k>k^*$, then $\mathrm{a}_{(k)}=\tau_k$ for all $k>k^*+j$, where $j:=\#\{k>k^*:\tau_k\le\tau_K+C\}<\infty$.*

Proof. At least k of the a_i (those with $i\le k$) are $\le\tau_k+C$, so $\mathrm{a}_{(k)}\le\tau_k+C$; at least k of them (those with $i\ge k$) are $\ge\tau_k-C$, hence at most $k-1$ are $<\tau_k-C$, so $\mathrm{a}_{(k)}\ge\tau_k-C$. For the second claim, the elements of the multiset not exceeding τ_K+C are the k^* moved ones and the j clock points τ_k, $k^*<k\le k^*+j$; the remaining elements are exactly τ_k, $k>k^*+j$, in increasing order. □

Lemma 3.18 (Lipschitz bound for the Jacobian)**.** *For all $x,x'\in\mathbb{R}^{k_m^*}$, $\|D\mathbb{F}_m(x)-D\mathbb{F}_m(x')\|_{\infty\to\infty}\le\mathbb{H}_m\|x-x'\|_\infty$, and*

$$\|\mathbb{F}_m(x)-\mathbb{F}_m(0)-\mathbb{J}_m x\|_\infty\le\tfrac12\mathbb{H}_m\|x\|_\infty^2. \tag{3.13}$$

Proof. $\mathbb{F}_m(x)_\varphi=\sum_{k\le k_m^*}\mathsf{c}_\varphi(\kappa_k^{(m,0)}+x_k)+\text{const}$, so $D^2\mathbb{F}_m(x)_\varphi$ is diagonal with entries $\partial^2_{x_k}\mathbb{F}_m(x)_\varphi=-\int\varphi(u)u^2\cos((\kappa_k+x_k)u)\,du$, of modulus at most $\int|\varphi|u^2$. The row φ of $D\mathbb{F}_m(x)-D\mathbb{F}_m(x')$ has absolute sum at most $\sum_{k\le k_m^*}\int|\varphi|u^2\,|x_k-x'_k|\le\mathbb{H}_m\|x-x'\|_\infty$. The Taylor bound follows by integrating $D\mathbb{F}_m$ along the segment. □

Theorem 3.19 (Certificate for one stage)**.** *Suppose $\mathbb{J}_m$ has a right inverse $\mathbb{Q}_m:\mathbb{R}^{\mathrm{t}_m}\to\mathbb{R}^{k_m^*}$, $\mathbb{J}_m\mathbb{Q}_m=I_{\mathrm{t}_m}$, with $\|\mathbb{Q}_m\|_{\infty\to\infty}\le\mathbb{L}_m$, and*

$$\mathsf{l}_m := 2\mathbb{L}_m^2\mathbb{H}_m\mathsf{h}_m<1. \tag{A_m}$$

Then the map $\mathbb{T}_m(y):=y-\mathbb{F}_m(\mathbb{Q}_m y)$ is a contraction of the ball $\{\|y\|_\infty\le2\mathsf{h}_m\}\subset\mathbb{R}^{\mathrm{t}_m}$ with constant l_m, and $x_*:=\mathbb{Q}_m y_*$, y_* its fixed point, satisfies

$$\mathbb{F}_m(x_*)=0, \qquad \|x_*\|_\infty\le2\mathbb{L}_m\mathsf{h}_m. \tag{3.14}$$

Proof. $D\mathbb{T}_m(y)=I-D\mathbb{F}_m(\mathbb{Q}_m y)\mathbb{Q}_m=(\mathbb{J}_m-D\mathbb{F}_m(\mathbb{Q}_m y))\mathbb{Q}_m$ because $\mathbb{J}_m\mathbb{Q}_m=I$, so by Lemma 3.18, for $\|y\|_\infty\le2\mathsf{h}_m$, $\|D\mathbb{T}_m(y)\|_{\infty\to\infty}\le\mathbb{H}_m\|\mathbb{Q}_m y\|_\infty\mathbb{L}_m\le2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m=\mathsf{l}_m<1$. Moreover $\mathbb{T}_m(y)=-\mathbb{F}_m(0)-(\mathbb{F}_m(\mathbb{Q}_m y)-\mathbb{F}_m(0)-\mathbb{J}_m\mathbb{Q}_m y)$, and by (3.13) the bracket has norm at most $\frac12\mathbb{H}_m\mathbb{L}_m^2\|y\|_\infty^2\le2\mathbb{H}_m\mathbb{L}_m^2\mathsf{h}_m^2$. Hence $\|\mathbb{T}_m(y)\|_\infty\le\mathsf{h}_m(1+\mathsf{l}_m)<2\mathsf{h}_m$: the ball is mapped into itself. Banach's fixed point theorem gives a unique fixed point y_* in the ball, $y_*=y_*-\mathbb{F}_m(\mathbb{Q}_m y_*)$, that is, $\mathbb{F}_m(x_*)=0$ with $\|x_*\|_\infty\le\mathbb{L}_m\|y_*\|_\infty\le2\mathbb{L}_m\mathsf{h}_m$. The iteration $y_{q+1}=\mathbb{T}_m(y_q)$, $y_0=0$, converges to y_* geometrically with ratio l_m; in the variable x it is $x_{q+1}=x_q-\mathbb{Q}_m\mathbb{F}_m(x_q)$, a Newton iteration with the frozen right inverse $\mathbb{Q}_m$. □

Theorem 3.20 (Staged certificate implies route B′)**.** *Let $C<\tau_1-2\pi$. Start at $\boldsymbol{\kappa}^{(1,0)}:=\boldsymbol{\tau}$ and, for $m\ge1$, let $\boldsymbol{\kappa}^{(m)}$ be the nondecreasing rearrangement of $\boldsymbol{\kappa}^{(m,0)}+x_*^{(m)}$ given by Theorem 3.19, and $\boldsymbol{\kappa}^{(m+1,0)}:=\boldsymbol{\kappa}^{(m)}$. Suppose (A_m) holds for every m and*

$$\sup_{m\ge1}\sum_{j\le m}2\mathbb{L}_j\mathsf{h}_j\le C. \tag{B}$$

Then every $\boldsymbol{\kappa}^{(m)}$ is C-admissible and equals the clock beyond a finite index $\widetilde{k}_m^*$, $\mathsf{s}_m(\boldsymbol{\kappa}^{(m)},\mathsf{a}^\zeta)=0$ for every m, $B'(C)$ holds, and therefore (Corollary 3.14) $\boldsymbol{\kappa}^{(m)}\to\boldsymbol{\gamma}$ coordinatewise and the Riemann hypothesis holds. In (A_{m+1}) and (B) the quantity h_{m+1} is $\max_{\varphi\in\mathscr{T}_{m+1}\setminus\mathscr{T}_m}|\langle\mathfrak{R}(\boldsymbol{\kappa}^{(m)},\mathsf{a}^\zeta),\varphi\rangle|$, the residual of the new tests only.

Proof. Induction on m. Suppose $\boldsymbol{\kappa}^{(m,0)}$ is C_{m-1}-admissible with $C_{m-1} := \sum_{j<m} 2\mathbb{L}_j \mathsf{h}_j$ ($C_0 = 0$) and equals the clock beyond a finite index. By Theorem 3.19, $\boldsymbol{\kappa}^{(m,0)} + x_*^{(m)}$ satisfies $|(\cdot)_k - \tau_k| \le C_{m-1} + 2\mathbb{L}_m \mathsf{h}_m = C_m$ for all k, and only finitely many coordinates differ from the clock. Lemma 3.17 shows that its rearrangement $\boldsymbol{\kappa}^{(m)}$ satisfies $|\kappa_k^{(m)} - \tau_k| \le C_m \le C$ and equals the clock beyond some $\widetilde{k}_m^*$; it is nondecreasing, and $\kappa_1^{(m)} \ge \tau_1 - C > 2\pi$. So $\boldsymbol{\kappa}^{(m)}$ is C-admissible. The residual $\mathfrak{R}$ depends only on the multiset of ordinates, so $\langle \mathfrak{R}(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta), \varphi \rangle = \mathbb{F}_m(x_*^{(m)})_\varphi = 0$ for all $\varphi \in \mathscr{T}_m$, that is, $\mathsf{s}_m = 0$. Since $\mathscr{T}_m \subset \mathscr{T}_{m+1}$, the start of stage $m+1$ has zero residual on $\mathscr{T}_m$, which gives the stated form of h_{m+1}. B$'(C)$ holds with $\mathsf{s}_m = 0$, and Corollary 3.14(a) gives the rest. □

Remark 3.21 (What the certificate theorem does and does not do). Theorem 3.20 replaces the extension lemma of the previous version (there Hypothesis E) by two computable quantities per stage, $\mathsf{l}_m = 2\mathbb{L}_m^2 \mathbb{H}_m \mathsf{h}_m$ and $2\mathbb{L}_m \mathsf{h}_m$. It requires no contraction factor q, no separate control of the perturbation of old tests, and no prior "entry into the basin": old tests are kept in $\mathscr{T}_{m+1}$ and are solved again exactly. The two conditions are not independent of the height: $\mathbb{H}_m \ge k_m^* \int |\varphi| u^2$ grows with the number of unknowns, and $\mathbb{L}_m$ is the ℓ^∞ norm of a right inverse of the Jacobian, whose growth is measured in Section 7. The conditions (A$_m$) for all m and (B) together imply the Riemann hypothesis, so they are not a weaker intermediate step; what changes is that each stage is a statement about one finite matrix and one finite residual vector, verifiable in principle by interval arithmetic. The restriction $C < \tau_1 - 2\pi = 8.24\ldots$ only ensures $\kappa_1 \ge 2\pi$ and can be removed by treating the first coordinate separately.

Proposition 3.22 (Graded extension)**.** *Let* $\mathsf{s}_m \ge 0$ *satisfy* $\mathsf{s}_{m+1} \le \mathsf{q}_m \mathsf{s}_m + \mathsf{h}_m$ *with* $0 \le \mathsf{q}_m \le 1$, $\prod_{m \ge 1} \mathsf{q}_m = 0$ *and* $\sum_m \mathsf{h}_m < \infty$. *Then* $\mathsf{s}_m \to 0$. *If the lists* $\boldsymbol{\kappa}^{(m)}$ *producing* $\mathsf{s}_m = \mathsf{s}_m(\boldsymbol{\kappa}^{(m)}, \mathsf{a}^\zeta)$ *are* C*-admissible sections for a fixed* C*, then* B$'(C)$ *holds.*

Proof. Iterating, $\mathsf{s}_m \le \mathsf{s}_1 \prod_{j<m} \mathsf{q}_j + \sum_{\ell<m} \mathsf{h}_\ell \prod_{\ell<j<m} \mathsf{q}_j$. Given $\epsilon > 0$ choose L with $\sum_{\ell \ge L} \mathsf{h}_\ell < \epsilon$. The terms with $\ell < L$ are at most $(\mathsf{s}_1 + \sum_{\ell<L} \mathsf{h}_\ell) \prod_{L \le j<m} \mathsf{q}_j \cdot \max(1, \ldots)$; more precisely each product $\prod_{\ell<j<m} \mathsf{q}_j$ with $\ell < L$ is at most $\prod_{L \le j<m} \mathsf{q}_j \to 0$ as $m \to \infty$, since all $\mathsf{q}_j \le 1$ and $\prod_j q_j = 0$. The terms with $\ell \ge L$ are at most $\sum_{\ell \ge L} \mathsf{h}_\ell < \epsilon$. Hence $\limsup \mathsf{s}_m \le \epsilon$. The second claim is Definition 3.13. □

Remark 3.23 (Tube budget, and the two geometries). In Proposition 3.22 the tube $|\kappa_k^{(m)} - \tau_k| \le C$, uniform in m, is a separate hypothesis: summability of the coordinate increments $\sum_m |\kappa_k^{(m+1)} - \kappa_k^{(m)}|$ gives convergence of each coordinate but not that the limit stays in a prescribed tube, so a construction must carry a displacement budget $\sum_{m \ge m(k)} |\kappa_k^{(m+1)} - \kappa_k^{(m)}| \le C - |\kappa_k^{(m(k))} - \tau_k|$. For B$'(C)$ itself no convergence of the whole sequence is needed: compactness in the product of the intervals $[\tau_k - C, \tau_k + C]$ (Proposition 3.4) extracts a convergent subsequence, and uniqueness identifies the limit. A second point concerns norms. Theorem 3.19 is stated in ℓ^∞, where $\mathbb{H}_m$ has the simple form (3.12) because the Hessian of each c_φ is diagonal. A bound $\|\mathbb{J}_m^\dagger\|_{2\to 2} \le 1/\sigma_{\min}(\mathbb{J}_m)$ is a Euclidean statement; passing to ℓ^∞ costs factors $\sqrt{\mathrm{t}_m}$ and $\sqrt{K_m}$ unless the right inverse is controlled directly in $\ell^\infty_{\mathrm{t}_m} \to \ell^\infty_{K_m}$. The numerical constants of Section 7 are computed directly in ℓ^∞.

Heuristic 3.24 (Frame bound from exponential sums of the clock)**.** In the opposite regime $k_m^* \gg \mathrm{t}_m$ one may take $\mathbb{Q}_m := \mathbb{J}_m^T (\mathbb{J}_m \mathbb{J}_m^T)^{-1}$. If $\|\frac{1}{K_m} \mathbb{J}_m \mathbb{J}_m^T - D\|_{\infty \to \infty} \le \epsilon < d_-$ for a diagonal $D \ge d_- I$, and $|(\mathbb{J}_m)_{\varphi,k}| \le \mathrm{a}_J$, then $\mathbb{L}_m \le \mathrm{t}_m A / (k_m^*(d_- - \epsilon)) = O(\mathrm{t}_m / k_m^*)$ and $\mathbb{L}_m^2 \mathbb{H}_m = O(\mathrm{t}_m^2 B / k_m^*)$ with $\mathrm{B}_0 := \max_\varphi \int |\varphi| u^2$, so (A$_m$) becomes $k_m^* \gg \mathrm{t}_m^2 B \mathsf{h}_m$: a wide block compensates a moderate residual. In the point model $\varphi \approx \delta_{u_j}$ the entries of $\frac{1}{k^*} \mathbb{J}\mathbb{J}^T$ are $u_j u_\ell \cdot \frac{1}{k^*} \sum_{k \in \mathfrak{B}} \sin(\tau_k u_j) \sin(\tau_k u_\ell)$, so the frame property reduces to cancellation in the exponential sums $\frac{1}{k^*} \sum_{k \in \mathfrak{B}} e^{i \mathrm{v} \tau_k}$ at the frequencies $\mathrm{v} = u_j \pm u_\ell \ne 0$. On a block of k^* consecutive clock points at height T the spacing is $\Delta_T \approx 2\pi / \log T$

with curvature $O((k^*)^2/(T\log^3 T))$, which suggests $|\frac{1}{k}^*\sum e^{iv\tau_k}| \lesssim \frac{\log T}{K|v|} + \frac{|v|(k^*)^2}{T\log^3 T}$ and a window $\frac{\log T}{h} \ll k^* \ll (T\log^3 T/u^*)^{1/2}$ for tests on a grid of step Δu. None of this is proved here, and the point model omits a decisive feature of smooth tests: a test of width w in u has $|\partial_\kappa \mathsf{c}_\varphi(\kappa)| \lesssim \mathrm{w}^{-1}e^{-(\mathrm{w}\kappa)^2/2}$ for Gaussian φ, so ordinates with $\mathrm{w}\kappa \gg 1$ are invisible to it and the corresponding columns of $\mathbb{J}_m$ are exponentially small. The underdetermined right inverse then has $\mathbb{L}_m$ dominated by the invisible columns; in the computations of Section 7 this produced $\mathbb{L}_m$ of size 10^{11} within four stages, and the stable regime was the opposite one, $\mathrm{t}_m \geq k^*_m$ with the unknowns restricted to the visible ordinates. The quasi-orthogonality heuristic is therefore a statement about tests that are not in $C_c^\infty(0,\infty)$, and it is not pursued further here.

3.3. **Explicit lists in the tube: Gonek's symmetrised Euler products.** Gonek [7] introduced, for $\mathrm{X} \geq 2$, the truncated Euler product $P_{\mathrm{X}}(s) := \exp(\sum_{n\leq \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{n^s \log n})$ with $\Lambda_{\mathrm{X}}(n) := \Lambda(n)$ for $n \leq \mathrm{X}$ and $\Lambda_{\mathrm{X}}(n) := \Lambda(n)\log(\mathrm{X}^2/n)/\log \mathrm{X}$ for $\mathrm{X} < n \leq \mathrm{X}^2$, and the symmetrised function $\zeta_{\mathrm{X}}(s) := P_{\mathrm{X}}(s) + \chi(s)P_{\mathrm{X}}(1-s)$, where χ is the factor of the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$. On the critical line $\chi(\frac12 + it) = e^{-2i\vartheta(t)}$ and $P_{\mathrm{X}}(\frac12 - it) = \overline{P_{\mathrm{X}}(\frac12 + it)}$, so with

$$\mathrm{p}_{\mathrm{X}}(t) := \arg P_{\mathrm{X}}(\tfrac12 + it) = -\sum_{n\leq \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)\sin(t\log n)}{\sqrt{n}\log n}, \tag{3.15}$$

and the two constants

$$\mathrm{m}_{\mathrm{X}} := \sum_{n\leq \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}\log n}, \qquad \mathrm{n}_{\mathrm{X}} := \sum_{n\leq \mathrm{X}^2} \frac{\Lambda_{\mathrm{X}}(n)}{\sqrt{n}}, \tag{3.16}$$

one has $\zeta_{\mathrm{X}}(\frac12 + it) = 2|P_{\mathrm{X}}(\frac12 + it)|\, e^{-i\vartheta(t)}\cos(\vartheta(t) + \mathrm{p}_{\mathrm{X}}(t))$. Since P_{X} has no zeros, the zeros of ζ_{X} on the line are exactly the solutions of

$$\Theta_{\mathrm{X}}(t) := \vartheta(t) + \mathrm{p}_{\mathrm{X}}(t) \in \tfrac{\pi}{2} + \pi\mathbb{Z}. \tag{3.17}$$

Gonek proves [7, Thm. 6.2 of the arXiv version] that every zero of ζ_{X} in $0 \leq \Re s \leq 1$ with imaginary part at least an absolute constant C_0 lies on $\Re s = \frac12$, unconditionally, and that under the Riemann hypothesis the zeros of ζ_{X} approach those of ζ as $\mathrm{X} \to \infty$. The phase (3.17) is the clock phase plus a truncated prime sum: $\mathcal{N}_{\mathrm{RVM}}(t) = 1 + \vartheta(t)/\pi + \mathrm{r}_\vartheta(t)$ with $|\mathrm{r}_\vartheta(t)| \leq \mathrm{c}_r/t$ for $t \geq 1$, by Stirling's formula [19, §12.33], so (3.17) reads $\mathcal{N}_{\mathrm{RVM}}(t) + \mathrm{p}_{\mathrm{X}}(t)/\pi \in \frac12 + \mathbb{Z} + \mathrm{r}_\vartheta(t)$, to be compared with $\mathcal{N}_{\mathrm{RVM}}(\tau_k) = k - \frac12$.

Lemma 3.25 (Gonek lists lie in a tube around the clock)**.** *Fix* $\mathrm{X} \geq 2$ *and put* $\mathrm{t}_{\mathrm{X}} := \max(2\pi e^{2\mathrm{n}_{\mathrm{X}}+2}, t_1)$*, where* t_1 *is such that* $|\vartheta'(t) - \frac12\log\frac{t}{2\pi}| \leq \frac12$ *for* $t \geq t_1$*. Then* Θ_{X} *is strictly increasing on* $[\mathrm{t}_{\mathrm{X}}, \infty)$ *with* $\Theta'_{\mathrm{X}} \geq \frac12$*, the solutions of* (3.17) *in* $[\mathrm{t}_{\mathrm{X}}, \infty)$ *form a strictly increasing sequence* $\mathrm{t}_1^{(\mathrm{X})} < \mathrm{t}_2^{(\mathrm{X})} < \cdots$*, each a simple zero of* ζ_{X}*, and there is an integer* j_{X} *such that the list*

$$\kappa_k^{(\mathrm{X})} := \begin{cases} \tau_k, & k \leq \mathrm{j}_{\mathrm{X}}, \\ \mathrm{t}_{k-\mathrm{j}_{\mathrm{X}}}^{(\mathrm{X})}, & k > \mathrm{j}_{\mathrm{X}}, \end{cases} \tag{3.18}$$

is nondecreasing and c_{X}*-admissible for some* $\mathrm{c}_{\mathrm{X}} < \infty$*, with*

$$|\kappa_k^{(\mathrm{X})} - \tau_k| \leq \frac{2\mathrm{m}_{\mathrm{X}} + o(1)}{\log(\tau_k/2\pi)} \qquad (k \to \infty). \tag{3.19}$$

Proof. $|\mathrm{p}'_{\mathrm{X}}(t)| = |\sum_{n\leq \mathrm{X}^2} \Lambda_{\mathrm{X}}(n) n^{-1/2}\cos(t\log n)| \leq \mathrm{n}_{\mathrm{X}}$ and $|\mathrm{p}_{\mathrm{X}}| \leq \mathrm{m}_{\mathrm{X}}$. For $t \geq \mathrm{t}_{\mathrm{X}}$, $\vartheta'(t) \geq \frac12\log\frac{t}{2\pi} - \frac12 \geq \mathrm{n}_{\mathrm{X}} + \frac12$, so $\Theta'_{\mathrm{X}} \geq \frac12$. Hence Θ_{X} is a strictly increasing bijection of $[\mathrm{t}_{\mathrm{X}}, \infty)$ onto $[\Theta_{\mathrm{X}}(\mathrm{t}_{\mathrm{X}}), \infty)$, and the solutions of (3.17) are the preimages of the points $\frac{\pi}{2} + \pi j$, $j \geq j_0$, where j_0 is the least integer with $\frac{\pi}{2} + \pi j_0 \geq \Theta_{\mathrm{X}}(\mathrm{t}_{\mathrm{X}})$; they are simple zeros because $\frac{d}{dt}\cos\Theta_{\mathrm{X}} = -\Theta'_{\mathrm{X}}\sin\Theta_{\mathrm{X}} \neq 0$

there. The i-th solution $\mathrm{t}_i^{(\mathrm{X})}$ satisfies $\Theta_{\mathrm{X}}(\mathrm{t}_i^{(\mathrm{X})}) = \frac{\pi}{2} + \pi(j_0 + i - 1)$; we set $\mathrm{j}_{\mathrm{X}} := j_0$, so that $k = \mathrm{j}_{\mathrm{X}} + i$ gives $\Theta_{\mathrm{X}}(\kappa_k^{(\mathrm{X})}) = \pi(k - \frac{1}{2})$. Since $\mathcal{N}_{\mathrm{RVM}} = 1 + \vartheta/\pi + \mathrm{r}_\vartheta$,

$$\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})}) = k + \tfrac{1}{2} - \frac{\mathrm{p}_{\mathrm{X}}(\kappa_k^{(\mathrm{X})})}{\pi} + \mathrm{r}_\vartheta(\kappa_k^{(\mathrm{X})}), \tag{3.20}$$

so, after replacing j_{X} by $\mathrm{j}_{\mathrm{X}} - 1$,

$$|\mathcal{N}_{\mathrm{RVM}}(\kappa_k^{(\mathrm{X})}) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| \le \frac{\mathrm{m}_{\mathrm{X}}}{\pi} + \frac{\mathrm{c}_r}{\kappa_k^{(\mathrm{X})}} \tag{3.21}$$

(the unit shift from the constant 1 is absorbed into the integer j_{X}, fixed once and for all; in the computations of Section 7 the offset was determined by matching). The right-hand side is bounded, and $\mathcal{N}_{\mathrm{RVM}}(2t) - \mathcal{N}_{\mathrm{RVM}}(t) \to \infty$, so $\kappa_k^{(\mathrm{X})}/\tau_k \to 1$. By the mean value theorem, $|\kappa_k^{(\mathrm{X})} - \tau_k| \le (\mathrm{m}_{\mathrm{X}}/\pi + o(1))/\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k)$ with s_k between $\kappa_k^{(\mathrm{X})}$ and τ_k, and $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}_k) = \frac{1}{2\pi}\log\frac{\tau_k}{2\pi}(1 + o(1))$. This is (3.19); it gives $|\kappa_k^{(\mathrm{X})} - \tau_k| \to 0$, so the displacements are bounded, the list is eventually nondecreasing, and finitely many initial indices (where the list is the clock) are harmless. If monotonicity of the concatenated list fails at the junction $k = \mathrm{j}_{\mathrm{X}}$, one replaces the prefix τ_k, $k \le \mathrm{j}_{\mathrm{X}}$, by the nondecreasing rearrangement of Lemma 3.17, which keeps the tube. □

Remark 3.26 (What the Gonek lists give and what remains). The lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ use the primes up to X^2 and the clock, and nothing about the zeros of ζ. Their prime side is built in: formally, $\sum_k \delta(t - \kappa_k^{(\mathrm{X})}) = \frac{\Theta'_{\mathrm{X}}(t)}{\pi}\big[1 + 2\sum_{j\ge1}(-1)^j\cos(2j\Theta_{\mathrm{X}}(t))\big]$ where $\Theta'_{\mathrm{X}} > 0$, and the $j = 0$ term carries $\mathrm{p}'_{\mathrm{X}}/\pi = -\frac{1}{\pi}\sum_{n\le \mathrm{X}^2}\Lambda_{\mathrm{X}}(n)n^{-1/2}\cos(t\log n)$, whose cosine transform is a sum of masses $\Lambda_{\mathrm{X}}(n)/(2\sqrt{n})$ at $u = \log n$, equal to a_n^ζ for $n \le \mathrm{X}$. Two statements would make them a solution of B′: (G1) a tube uniform in X, $\sup_{\mathrm{X}\ge\mathrm{X}_0}\sup_k|\kappa_k^{(\mathrm{X})} - \tau_k| \le C$, which by (3.19) concerns the size of p_{X} at its own crossing points and not its supremum ($\mathrm{m}_{\mathrm{X}} \asymp \mathrm{X}/\log\mathrm{X}$ is useless for this); and (G2) $\langle\mathfrak{R}(\boldsymbol{\kappa}^{(\mathrm{X})}, \mathsf{a}^\zeta), \varphi\rangle \to 0$ as $\mathrm{X} \to \infty$ for each test φ, whose content is the vanishing of the harmonics $j \ge 1$ against φ. Neither is proved here. The monotonicity threshold $\mathfrak{t}_{\mathrm{X}}$ of Lemma 3.25 grows like $e^{2\mathrm{n}_{\mathrm{X}}}$, so the global construction cannot rely on monotonicity of Θ_{X} at fixed height as $\mathrm{X} \to \infty$; Gonek's theorem on the location of the zeros does not need it. For fixed X the tube constant in (3.19) is $2\mathrm{m}_{\mathrm{X}}$: $\mathrm{m}_3 = 1.42$, $\mathrm{m}_8 = 3.02$, $\mathrm{m}_{20} = 5.98$. Numerically (Section 7) the lists with $3 \le \mathrm{X} \le 20$ lie within 0.19 to 0.035 mean spacings of the zeta ordinates for $k \le 400$, whereas the clock is within 0.66 spacings.

4. An a priori bound for quasi-solutions

Throughout this section $(\boldsymbol{\kappa}, \mathsf{a})$ is a finite section in $\mathbb{K}(C, \mathbb{M})$ with parameters k^*, n^*.

Definition 4.1 (Laplace residual)**.** For $\Re s > 1$ put

$$\widehat{\mathsf{R}}(s) := \tfrac{1}{2}\frac{P'_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\kappa}}}(s) - \tfrac{1}{2}\frac{G'}{G}(s) + \sum_{n=2}^{n^*}\mathsf{a}_n\, n^{\frac{1}{2}-s}. \tag{4.1}$$

For $T \ge 0$ put

$$\mathbb{E}(T) := \int_{-T}^{T}\big|\widehat{\mathsf{R}}(2 + iv)\big|\,dv, \qquad \upsilon := \sup_{T\ge2}\frac{\mathbb{E}(T)}{\log T} \in [0, \infty]. \tag{4.2}$$

Since $\int_0^\infty e^{-(s-\frac12)u}\cos(\kappa u)\,du = \frac{s-\frac12}{(s-\frac12)^2+\kappa^2} = \frac12\frac{d}{ds}\log(1 + \frac{q(s)}{\frac14+\kappa^2})$ and $\int_0^\infty e^{-(s-\frac12)u}\delta(u - \log n)\,du = n^{\frac12-s}$, $\widehat{\mathsf{R}}$ is formally the Laplace transform of the residual distribution of (2.4), with the normalisation of k fixed in `eq:transport-defs`; this is made precise in Lemma 3.10. We take (4.1) as the definition. Because the coefficients are real, $\widehat{\mathsf{R}}(\bar s) = \overline{\widehat{\mathsf{R}}(s)}$, so $\mathbb{E}(T) = 2\int_0^T|\widehat{\mathsf{R}}(2 + iv)|\,dv$. We assume $\upsilon < \infty$ throughout this section; if $\upsilon = \infty$ every bound below is void. By Lemma 3.10, $\widehat{\mathsf{R}}$ is the Laplace transform of the residual distribution of the section.

Constants. Put $\mathrm{c}_\ell := 2\sum_{n\ge2} n^{-3/2}/\log n < \infty$, and, for $s \in \mathbb{C}$ and real $x \ge 2\pi$, $\mathfrak{f}(x;s) := \log(1+\frac{q(s)}{\frac14+x^2})$, with the principal branch, which is defined whenever $|s-\frac12| < x$. Let $\mathbb{J} := \{2\}\cup\{\sigma+i : -1 \le \sigma \le 2\}$, a compact set on which $|s-\frac12| \le \frac52 < 2\pi$, and

$$\mathfrak{B}(C) := \sum_{k\ge1} \sup_{x\in[\max(\tau_k-C,2\pi),\,\tau_k+C]} \sup_{s\in\mathbb{J}} |\mathfrak{f}(x;s) - \mathfrak{f}(\tau_k;s)|. \tag{4.3}$$

This is finite. On $\mathbb{J}$, $|q(s)| \le 5$ and $|\partial_x f(x;s)| = |\frac{2xq(s)}{(\frac14+x^2)(x^2+(s-\frac12)^2)}| \le \frac{20}{x^3}$ for $x \ge 4$. The k-th term is therefore at most $160C\tau_k^{-3}$ when $\tau_k \ge 2C$, and $\sum_k \tau_k^{-3} < \infty$. Further, $\mathrm{c}_\tau := \max_{-1\le\sigma\le2} |\log|\zeta_{\mathrm{RVM}}(\sigma+i)||$ and $\mathrm{c}_{\mathrm{RVM}} := \sup_{\Re s\ge2} |\log\zeta_{\mathrm{RVM}}(s)|$. Both are finite: the first because ζ_{RVM} has no zeros or poles on that segment, the second by `thm:halfplane-RVM`.

Lemma 4.2 (Phragmén–Lindelöf, two forms)**.** (a) *Let $\mathbb{S} := \{\mathrm{l}_1 \le \Re s \le \mathrm{l}_2,\ \Im s \ge t_1\}$ have width $a := \mathrm{l}_2 - \mathrm{l}_1$. Let u be upper semicontinuous on $\mathbb{S}$ and subharmonic in its interior, with $u \le \mathrm{M}_0$ on $\partial\mathbb{S}$ and $u(s) \le \mathrm{A}_0 e^{c\Im s}$ for some $c < \pi/a$. Then $u \le \mathrm{M}_0$ on $\mathbb{S}$.*
(b) *Let $\mathfrak{g}$ be holomorphic in $\Re s > 1$, continuous on $\Re s \ge 2$, with $|g| \le 1$ on $\Re s = 2$ and $|\mathfrak{g}(s)| \le \mathrm{A}_0(2+|s|)^{\mathrm{L}}$ on $\Re s \ge 2$. Then $|g| \le 1$ on $\Re s \ge 2$.*

Proof. (a) Choose $c < c' < \pi/a$ and $m := \frac{\mathrm{l}_1+\mathrm{l}_2}{2}$, and put $\omega(s) := e^{c'\Im s}\cos(c'(\Re s - m))$. This is harmonic, being the real part of $e^{-ic'(s-m)}$. On $\mathbb{S}$, $\omega \ge \cos(c'a/2)\, e^{c'\Im s}$, and $\cos(c'a/2) > 0$. For $\mathrm{p} > 0$, $u - \mathrm{M}_0 - \mathrm{p}\omega$ is subharmonic, ≤ 0 on $\partial\mathbb{S}$, and tends to $-\infty$ as $\Im s \to \infty$. The maximum principle on $\mathbb{S}\cap\{\Im s \le t_2\}$, with t_2 large, gives $u \le \mathrm{M}_0 + \mathrm{p}\omega$. Let $\mathrm{p} \to 0$.
(b) For $\mathrm{p} > 0$, $\mathfrak{g}_\eta(s) := \mathfrak{g}(s)\exp(-\mathrm{p}(s-1)^{1/2})$, with the principal branch, satisfies $|\mathfrak{g}_\eta| \le |g|$ on $\Re s \ge 2$, because $\Re(s-1)^{1/2} \ge \cos(\pi/4)|s-1|^{1/2} > 0$ there. Moreover $\mathfrak{g}_\eta(s) \to 0$ as $|s| \to \infty$ in $\Re s \ge 2$. The maximum principle on large half-discs gives $|\mathfrak{g}_\eta| \le 1$. Let $\mathrm{p} \to 0$. □

Lemma 4.3 (The Dirichlet factor)**.** *Put $\mathcal{D}_{\mathsf{a}}(s) := \sum_{n=2}^{n^*} \frac{2\mathsf{a}_n}{\log n} n^{\frac12 - s}$ and $\mathcal{Y} := \zeta_{\boldsymbol{\kappa}} e^{-\mathcal{D}_{\mathsf{a}}}$. Then $\mathcal{Y}$ is holomorphic and zero-free in $\Re s > 1$, and $|\mathcal{D}_{\mathsf{a}}| \le \mathbb{M}\mathrm{c}_\ell$ on $\Re s \ge 2$. For the branch of $\log\mathcal{Y}$ that is real on $(1,\infty)$,*

$$(\log\mathcal{Y})' = 2\,\widehat{\mathsf{R}}, \qquad |\log\mathcal{Y}(2)| \le \mathbb{B}_0 := |\log\zeta_{\mathrm{RVM}}(2)| + \mathfrak{B}(C) + \mathbb{M}\mathrm{c}_\ell, \tag{4.4}$$

and, for all real t,

$$|\log\mathcal{Y}(2+it)| \le \mathbb{B}_0 + \mathbb{E}(|t|) \le \mathbb{B}_0 + \upsilon\log(2+|t|). \tag{4.5}$$

Proof. $P_{\boldsymbol{\kappa}}$ vanishes only on $\Re s = \frac12$, and G is holomorphic and zero-free in $\Re s > 1$. Hence $\zeta_{\boldsymbol{\kappa}}$ and $\mathcal{Y}$ are holomorphic and zero-free there. $|\mathcal{D}_{\mathsf{a}}(s)| \le \sum_n 2Mn^{\frac12-\Re s}/\log n \le \mathbb{M}\mathrm{c}_\ell$ for $\Re s \ge 2$. Differentiating, $(\log\mathcal{Y})' = \frac{P'_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\kappa}}} - \frac{G'}{G} + 2\sum_n \mathsf{a}_n n^{\frac12-s} = 2\widehat{\mathsf{R}}$. At $s = 2$, $\log\zeta_{\boldsymbol{\kappa}}(2) = \log\zeta_{\mathrm{RVM}}(2) + \sum_{k\le k^*}(\mathfrak{f}(\kappa_k;2) - \mathfrak{f}(\tau_k;2))$, a finite sum of real numbers bounded by $\mathfrak{B}(C)$, and $|\mathcal{D}_{\mathsf{a}}(2)| \le \mathbb{M}\mathrm{c}_\ell$. Finally, $\log\mathcal{Y}(2+it) - \log\mathcal{Y}(2) = 2i\int_0^t \widehat{\mathsf{R}}(2+iv)\,dv$ has modulus at most $\mathbb{E}(|t|)$. For $|t| \ge 2$, $\mathbb{E}(|t|) \le \upsilon\log|t|$; for $|t| < 2$, $\mathbb{E}(|t|) \le \mathbb{E}(2) \le \upsilon\log 2$. □

Lemma 4.4 (Bound on the half-plane)**.** *With $\mathbb{W} := \log|\zeta_{\boldsymbol{\kappa}}| = \log|P_{\boldsymbol{\kappa}}| - \log|G|$ and $\mathbb{B}_1 := \mathbb{B}_0 + \upsilon\log(2\sqrt2)$, $\mathbb{B}_2 := \mathbb{B}_1 + \mathbb{M}\mathrm{c}_\ell$, for $\Re s \ge 2$,*

$$|\log|\mathcal{Y}(s)|| \le \mathbb{B}_1 + \upsilon\log|s-1|, \qquad |\mathbb{W}(s)| \le \mathbb{B}_2 + \upsilon\log|s-1|. \tag{4.6}$$

Proof. On $\Re s = 2$, $|s-1| = \sqrt{1+t^2} \ge (2+|t|)/(2\sqrt2)$, so (4.5) gives $|\mathcal{Y}(s)|^{\pm1} \le e^{B_1}|s-1|^{\upsilon}$. A priori growth: on $\Re s \ge 2$, $|\log|\zeta_{\mathrm{RVM}}|| \le \mathrm{c}_{\mathrm{RVM}}$. The quotient $\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}$ is a finite product of k^* factors $\frac{(\kappa_k^2+w^2)(\frac14+\tau_k^2)}{(\frac14+\kappa_k^2)(\tau_k^2+w^2)}$ with $w = s-\frac12$. Since $|\kappa^2+w^2| = |\kappa - \Im s + i\Re w|\,|\kappa + \Im s - i\Re w| \ge (\Re w)^2 \ge \frac94$, each factor and its inverse are $O((2+|s|)^4)$. Hence $|\mathcal{Y}|^{\pm1} \le \mathrm{A}_0(2+|s|)^{4k^*}$. Apply Lemma 4.2(b) to $\mathfrak{g}_+ := \mathcal{Y}\,(s-1)^{-\upsilon}e^{-\mathbb{B}_1}$ and to $\mathfrak{g}_- := \mathcal{Y}^{-1}(s-1)^{-\upsilon}e^{-\mathbb{B}_1}$, with the principal branch of $(s-1)^{-\upsilon}$,

holomorphic in $\Re s > 1$ with $|(s-1)^{-\upsilon}| = |s-1|^{-\upsilon}$; both are holomorphic in $\Re s > 1$ and of polynomial growth on $\Re s \geq 2$. Finally $\mathbb{W} = \log|\mathcal{Y}| + \Re\mathcal{D}_{\mathfrak{a}}$. □

Lemma 4.5 (Bound on the half-strip)**.** *Let* $\mathbb{S} := \{-1 \leq \Re s \leq 2,\ \Im s \geq 1\}$ *and* $\mathbb{B}_3 := \max\big(\mathbb{B}_2 + \frac{3}{4}\log 2,\ \mathrm{c}_\tau + \mathfrak{B}(C)\big)$. *Then*

$$\mathbb{W}(s) \leq \mathbb{B}_3 + \big(\tfrac{3}{2} + \upsilon\big)\log|s+2| \qquad (s \in \mathbb{S}). \tag{4.7}$$

Proof. $\mathbb{W}$ is subharmonic in the interior of $\mathbb{S}$, since $\log|G|$ is harmonic there (G has no zeros or poles with $\Im s \geq 1$), and it is upper semicontinuous on $\mathbb{S}$. The majorant $\mathrm{h}_0(s) := (\frac{3}{2} + \upsilon)\log|s+2|$ is harmonic and nonnegative on $\mathbb{S}$. We check the three sides.

Right side. By (4.6), $\mathbb{W}(2+it) \leq \mathbb{B}_2 + \upsilon\log|1+it| \leq \mathbb{B}_2 + \mathrm{h}_0(2+it)$.

Left side. $P_{\boldsymbol{\kappa}}$ and G are real, so $\mathbb{W}(\bar{s}) = \mathbb{W}(s)$. Combined with $P_{\boldsymbol{\kappa}}(1-s) = P_{\boldsymbol{\kappa}}(s)$, this gives $\mathbb{W}(-1+it) = \mathbb{W}(2+it) + \log|G(2+it)/G(-1+it)|$. By $\Gamma(z+1) = z\Gamma(z)$ and the reflection formula, $|\Gamma(\frac{1}{2}+iy)|^2 = \pi/\cosh\pi y$ and $|\Gamma(1+iy)|^2 = \pi y/\sinh\pi y$, so

$$\Big|\frac{G(2+it)}{G(-1+it)}\Big| = \frac{|1+it|}{2}\,\pi^{-3/2}\Big(\frac{t}{2}\coth\frac{\pi t}{2}\Big)^{1/2} \leq (1+t)^{3/2} \qquad (t \geq 1). \tag{4.8}$$

For the inequality: $|1+it| \leq 1+t$, $\frac{t}{2}\coth\frac{\pi t}{2} \leq \frac{1+t}{2}\coth\frac{\pi}{2}$, and $\frac{1}{2}\pi^{-3/2}(\frac{1}{2}\coth\frac{\pi}{2})^{1/2} < 1$. Since $(1+t) \leq \sqrt{2}|1+it|$ and $|s+2| = |1+it|$ at $s = -1+it$, we get $\mathbb{W}(-1+it) \leq \mathbb{B}_2 + \frac{3}{4}\log 2 + \mathrm{h}_0(-1+it)$.

Bottom side. For $s = \sigma + i$, $\mathbb{W}(s) = \log|\zeta_{\mathrm{RVM}}(s)| + \sum_{k\leq k^*}\Re(\mathfrak{f}(\kappa_k; s) - \mathfrak{f}(\tau_k; s)) \leq \mathrm{c}_\tau + \mathfrak{B}(C) \leq \mathrm{c}_\tau + \mathfrak{B}(C) + \mathrm{h}_0(s)$.

Growth. $\log|P_{\boldsymbol{\kappa}}(s)| \leq \sum_k \log(1 + |q(s)|/\kappa_k^2)$. Split at $\kappa_k \leq 2|s|$ and $\kappa_k > 2|s|$: the first part has $N_{\boldsymbol{\kappa}}(2|s|) = O(|s|\log|s|)$ terms, each at most $\log(1+|q(s)|/4\pi^2) = O(\log|s|)$; in the second, $\log(1+|q(s)|/\kappa_k^2) \leq |q(s)|\kappa_k^{-2}$, and $\sum_{\kappa_k > 2|s|}\kappa_k^{-2} = O(\log|s|/|s|)$ by partial summation with $N_{\boldsymbol{\kappa}}(t) = O(t\log t)$. Hence $\log|P_{\boldsymbol{\kappa}}(s)| = O(|s|\log^2|s|)$, and $-\log|G(s)| \leq \frac{\pi}{4}|\Im s| + O(\log|s|)$ on $\mathbb{S}$ by Stirling's formula [19, §12.33]. Hence $\mathbb{W} - \mathrm{h}_0 \leq \mathrm{A}_0 e^{c\Im s}$ for any $c > 0$. Lemma 4.2(a), with width 3 and $c < \pi/3$, applied to $\mathbb{W} - \mathrm{h}_0$ gives the claim. □

Lemma 4.6 (Backlund–Jensen on a horizontal segment)**.** *There are absolute constants* $T_0 \geq 3$ *and* c_5 *such that, for* $T \geq T_0$ *with* $T \notin \{\kappa_k\}$, *the variation of* $\arg P_{\boldsymbol{\kappa}}$ *along the segment from* $2+iT$ *to* $\frac{1}{2}+iT$ *is at most* $\pi(n_T+1)$, *where*

$$n_T \leq \frac{(\frac{5}{2}+\upsilon)\log(T+8) + \mathbb{E}(T) + \mathbb{B}_0 + \mathbb{B}_3 + \mathbb{M}\mathrm{c}_\ell + 2\mathrm{c}_5}{\log(4/3)}. \tag{4.9}$$

Proof. Choose $\mathrm{a}_T \in \mathbb{R}$ with $e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(2+iT) > 0$ and put $\mathfrak{F}_T(z) := \frac{1}{2}(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(z+iT) + e^{-i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(z-iT))$, which is entire. Since $P_{\boldsymbol{\kappa}}$ is real, $P_{\boldsymbol{\kappa}}(x-iT) = \overline{P_{\boldsymbol{\kappa}}(x+iT)}$ for real x, so $\mathfrak{F}_T(x) = \Re(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}}(x+iT))$ and $\mathfrak{F}_T(2) = |P_{\boldsymbol{\kappa}}(2+iT)|$. If $\mathfrak{F}_T$ has n_T zeros on $[\frac{1}{2}, 2]$, the segment splits into at most n_T+1 pieces on which $\Re(e^{i\mathrm{a}_T}P_{\boldsymbol{\kappa}})$ has constant sign. On each piece arg varies by less than π, which gives the first claim [17, §9.4].

The segment $[\frac{1}{2}, 2]$ lies in the closed disc $|z-2| \leq \frac{3}{2}$. For $\frac{3}{2} < r < 2$, Jensen's formula on $|z-2| \leq R = 2$ gives $n_T\log\frac{R}{r} \leq \max_{|z-2|=2}\log|\mathfrak{F}_T(z)| - \log|\mathfrak{F}_T(2)|$, because the zeros in the closed disc of radius r, which include the n_T zeros on the segment, contribute at least $n_T\log(R/r)$; letting $r \downarrow \frac{3}{2}$ gives the inequality with $\log\frac{4}{3}$. Moreover $|\mathfrak{F}_T(z)| \leq \max_{|\zeta-2|\leq 2}|P_{\boldsymbol{\kappa}}(\zeta+iT)|$, because $|P_{\boldsymbol{\kappa}}(z-iT)| = |P_{\boldsymbol{\kappa}}(\bar{z}+iT)|$. Write $\log|P_{\boldsymbol{\kappa}}| = \mathbb{W} + \log|G|$. On the disc $\{|s-(2+iT)| \leq 2\}$, the part with $\Re s \leq 2$ lies in $\mathbb{S}$ and $|s+2| \leq T+6$, so Lemma 4.5 gives $\mathbb{W} \leq \mathbb{B}_3 + (\frac{3}{2}+\upsilon)\log(T+6)$. The part with $\Re s \geq 2$ satisfies $\mathbb{W} \leq \mathbb{B}_2 + \upsilon\log(T+5)$ by Lemma 4.4, a smaller bound. At the centre, (4.5) gives $\mathbb{W}(2+iT) \geq -(\mathbb{B}_0 + \mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell)$. For the oscillation of $\log|G|$ on the disc: by Stirling's formula for Γ'/Γ in $|\arg z| \leq \frac{3}{4}\pi$ [19, §12.33], $|G'/G(s)| = |\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\frac{s}{2})| \leq \frac{1}{2}\log(T+8) + \mathrm{c}_5$ on the disc for $T \geq T_0$. So the oscillation is at most $\log(T+8) + 2\mathrm{c}_5$. Collecting terms gives (4.9). □

Lemma 4.7 (Argument principle)**.** *For $T > 0$, $T \notin \{\kappa_k\}$, let $\mathbb{G}_T$ be the path $2 \to 2+iT \to \frac{1}{2}+iT$. Then*

$$\pi N_{\boldsymbol{\kappa}}(T) = \Delta_{\mathbb{G}_T} \arg P_{\boldsymbol{\kappa}} = \pi + \vartheta(T) + \Delta_{\mathbb{G}_T} \arg \zeta_{\boldsymbol{\kappa}}, \tag{4.10}$$

where $\vartheta(T) = \arg\big(\pi^{-iT/2}\Gamma(\frac{1}{4}+\frac{iT}{2})\big)$ is the Riemann–Siegel theta function, defined by continuous variation from $\vartheta(0) = 0$.

Proof. Take the rectangle $\mathfrak{R}_T$ with vertices $2 \pm iT$, $-1 \pm iT$. $P_{\boldsymbol{\kappa}}$ has no zeros on $\partial\mathfrak{R}_T$: its zeros lie on $\Re s = \frac{1}{2}$, and $T \notin \{\kappa_k\}$, so $\frac{1}{2} \pm iT$ are not zeros. Its zeros inside $\mathfrak{R}_T$ are $\frac{1}{2} \pm i\kappa_k$ with $\kappa_k < T$, counted with multiplicity (coincident $\kappa_j = \kappa_k$ give a multiple zero), $2N_{\boldsymbol{\kappa}}(T)$ in total. Reflection $s \mapsto 1 - \bar{s}$ maps $\mathbb{G}_T$ to the path $-1 \to -1+iT \to \frac{1}{2}+iT$. Since $P_{\boldsymbol{\kappa}}(1-\bar{s}) = \overline{P_{\boldsymbol{\kappa}}(s)}$, the variation of $\arg P_{\boldsymbol{\kappa}}$ along the reversed image path $\frac{1}{2}+iT \to -1+iT \to -1$ equals $\Delta_{\mathbb{G}_T} \arg P_{\boldsymbol{\kappa}}$. The two lower quarters are handled in the same way with $s \mapsto \bar{s}$. The four quarter paths make up $\partial\mathfrak{R}_T$, so $4\Delta_{\mathbb{G}_T} \arg P_{\boldsymbol{\kappa}} = 2\pi \cdot 2N_{\boldsymbol{\kappa}}(T)$.

For the second equality, write $P_{\boldsymbol{\kappa}} = G\zeta_{\boldsymbol{\kappa}}$. Along $\mathbb{G}_T$, $s(s-1)$ moves from 2 to $-(\frac{1}{4}+T^2)$ inside the closed upper half-plane, so its argument varies by π. $\pi^{-s/2}\Gamma(s/2)$ is holomorphic and zero-free in $\Re s > 0$, so its argument variation along $\mathbb{G}_T$ equals that along $[2, \frac{1}{2}] \cup [\frac{1}{2}, \frac{1}{2}+iT]$, which is $0 + \vartheta(T)$. □

Theorem 4.8 (Counting bound for quasi-solutions)**.** *There are absolute constants T_0 and c_5 such that, for every finite section in $\mathbb{K}(C, \mathbb{M})$ and every $T \geq T_0$,*

$$\big|N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T)\big| \leq (\mathrm{c}_1 + \mathrm{c}_2\upsilon)\log(T+8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_5, \tag{4.11}$$

with

$$\mathrm{c}_1 = \frac{5}{2\log(4/3)} = 8.690\ldots, \quad \mathrm{c}_2 = \frac{2}{\log(4/3)} + \frac{1}{\pi} = 7.270\ldots, \quad \mathrm{c}_3 = \frac{1}{\log(4/3)} + \frac{1}{\pi} = 3.794\ldots, \tag{4.12}$$

and $\mathbb{B} := \mathbb{B}_0 + \mathbb{B}_3 + \mathbb{M}\mathrm{c}_\ell$. The constants $\mathrm{c}_1, \ldots, \mathrm{c}_5$ do not depend on k^*, n^*, C, $\mathbb{M}$ or υ; $\mathbb{B}$ depends on C (through $\mathfrak{B}(C)$), on $\mathbb{M}$, and linearly on υ (through $\mathbb{B}_2$), so the total coefficient of υ in (4.11) *is $\mathrm{c}_2\log(T+8) + \mathrm{c}_3\log(2\sqrt{2})$. In Corollary 4.9 the parameters C, $\mathbb{M}$, υ are fixed before $k \to \infty$.*

Proof. By continuity from the right it suffices to take $T \notin \{\kappa_k\}$. By Lemma 4.7, $N_{\boldsymbol{\kappa}}(T) - 1 - \vartheta(T)/\pi = \pi^{-1}\Delta_{\mathbb{G}_T} \arg \zeta_{\boldsymbol{\kappa}}$.

Vertical piece. $\arg \zeta_{\boldsymbol{\kappa}} = \Im \log \mathcal{Y} + \Im \mathcal{D}_{\mathsf{a}}$, with $\Im \log \mathcal{Y}(2) = \Im \mathcal{D}_{\mathsf{a}}(2) = 0$. Its variation is at most $\mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell$ by Lemma 4.3.

Horizontal piece. The variation is $\Delta \arg P_{\boldsymbol{\kappa}} - \Delta \arg G$. The first is at most $\pi(n_T+1)$ by Lemma 4.6. The second is $-\Im \int_{1/2}^{2} (G'/G)(\sigma + iT)\, d\sigma$. Here $|\Im(G'/G)| \leq |\frac{1}{s}| + |\frac{1}{s-1}| + \frac{1}{2}|\arg \frac{s}{2}| + O(1/T) \leq 1$ for $T \geq T_0$, by Stirling's formula for Γ'/Γ [19, §12.33]. So the second variation is at most $\mathrm{c}_6 := \frac{3}{2}$.

Main term. By the Stirling expansion of ϑ [19, §12.33], $1 + \vartheta(T)/\pi = \mathcal{N}_{\mathrm{RVM}}(T) + O(1/T)$, and this error is at most $1/T$ for $T \geq T_0$. With (2.2):

$$\big|N_{\boldsymbol{\kappa}}(T) - n_{\mathrm{RVM}}(T)\big| \leq n_T + 1 + \frac{\mathbb{E}(T) + \mathbb{M}\mathrm{c}_\ell + \mathrm{c}_6}{\pi} + \frac{1}{T} + \frac{1}{2}. \tag{4.13}$$

Insert (4.9) and $\mathbb{E}(T) \leq \upsilon\log(T+8)$. □

Corollary 4.9 (Displacement bound)**.** *Assume $T_0 \geq 2\pi e$. For every k with $\tau_k - C \geq T_0$,*

$$|\kappa_k - \tau_k| \leq \frac{2\pi\big[(\mathrm{c}_1 + \mathrm{c}_2\upsilon)\log(\tau_k + C + 8) + \mathrm{c}_3\mathbb{B} + \mathrm{c}_5\big]}{\log\big((\tau_k - C)/2\pi\big)}. \tag{4.14}$$

In particular $\limsup_{k\to\infty} |\kappa_k - \tau_k| \leq 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon)$ uniformly over sections with given $C, \mathbb{M}, \upsilon$. For every $\mathrm{p} > 0$ there is $k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ such that $|\kappa_k - \tau_k| \leq 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$ for $k \geq k_0$.

Proof. Let $\mathbb{D}(T)$ be the right-hand side of (4.11); it is nondecreasing. Suppose $\kappa_k > \tau_k + \mathrm{q}$ with $0 < d \le C$, and let $T := \tau_k + \mathrm{q} - \mathrm{h}'$ with small $\mathrm{h}' > 0$. The list is nondecreasing, so $N_{\boldsymbol{\kappa}}(T) \le k-1$. By (2.2) and because $\mathcal{N}'_{\mathrm{RVM}}$ is increasing, $n_{\mathrm{RVM}}(T) \ge \mathcal{N}_{\mathrm{RVM}}(T) - \frac{1}{2} \ge k - 1 + (\mathrm{q}-\mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k)$. Hence $(\mathrm{q}-\mathrm{h}')\frac{1}{2\pi}\log\frac{\tau_k}{2\pi} \le \mathbb{D}(\tau_k + C)$. Now suppose $\kappa_k < \tau_k - \mathrm{q}$, and let $T := \tau_k - \mathrm{q} + \mathrm{h}'$. Then $N_{\boldsymbol{\kappa}}(T) \ge k$ and $n_{\mathrm{RVM}}(T) \le \mathcal{N}_{\mathrm{RVM}}(T) + \frac{1}{2} \le k - (\mathrm{q}-\mathrm{h}')\mathcal{N}'_{\mathrm{RVM}}(\tau_k - C)$. Let $\mathrm{h}' \to 0$ in both cases. □

Remark 4.10 (What Corollary 4.9 does not give). The proof uses the imposed tube $|\kappa_k - \tau_k| \le C$ twice: through $\mathfrak{B}(C)$ in $\mathbb{B}$, and to replace $\mathbb{D}(\tau_k + d)$ by $\mathbb{D}(\tau_k + C)$. The conclusion is that, inside any finite tube, the coordinates of sufficiently high index lie in the smaller tube of radius $2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. So the constraint is inactive there once $C > 2\pi(\mathrm{c}_1 + \mathrm{c}_2\upsilon) + \mathrm{p}$. For the indices $k < k_0(C, \mathbb{M}, \upsilon, \mathrm{p})$ nothing is proved, and the corollary does not show that unconstrained near-minimisers lie in a universal tube. The proof uses finite sections: Lemma 4.4 needs the polynomial growth $\mathrm{A}_0(2+|s|)^{4k^*}$. Extending it to infinite lists requires a separate growth bound for $P_{\boldsymbol{\kappa}}/P_{\boldsymbol{\tau}}$ on $\Re s \ge 2$, which is not given here. The constant $2\pi\mathrm{c}_1 = 54.60\ldots$ is far from optimal.

Heuristic 4.11 (What a finite section controls)**.** The quantity in Theorem 4.8 is $\mathbb{E}(T)$, the L^1 norm on $\Re s = 2$ of the Laplace transform of the residual. It is not the Euclidean residual that a numerical section minimises over finitely many tests. Let the tests be $\mathrm{j}_0(u)\varphi(u)$ with a fixed window $\mathrm{j}_0 \in C_c^\infty(0,\infty)$, $\mathrm{j}_0 \equiv 1$ on $[u_0, u^*]$, and φ band-limited to $|\omega| \le \mathrm{w}^\sharp$. Then the kernel $\mathrm{j}_0(u)e^{-\frac{3}{2}u}e^{-ivu}$ is, approximately, in the span of the tests only for $|v| \lesssim \mathrm{w}^\sharp$. The residual on $(0, u_0)$ is then not controlled, and for a list at distance C from the clock it is expected to be of size $C\log(1/u)$ as $u \to 0$. Its contribution to $\mathbb{E}(T)$ is then $O(CTu_0\log(1/u_0))$. Since $u_0 \gtrsim 1/\mathrm{w}^\sharp$ in a band-limited section, Theorem 4.8 would apply with a given υ_0 only at heights $T \lesssim \upsilon_0\mathrm{w}^\sharp/(C\log\mathrm{w}^\sharp)$. None of these size estimates is proved here. They indicate which norm a section must make small. Lemma 4.3 shows that this norm is the right one.

Heuristic 4.12 (Informative part of a section)**.** Where the atoms are denser than the resolution of the tests, $e^{-u} \ll \pi/\mathrm{w}^\sharp$, the masses alone can approximate a smooth target by a Riemann sum $\mathsf{a}_n \approx \mathfrak{g}(\log n)/n$ without moving any ordinate. That part of the window then carries no information on the ordinates. Counting unknowns against equations in the remaining window, $u^* \le \log\mathrm{w}^\sharp + O(1)$, gives an overdetermined system. This proves nothing about compatibility. It only explains why B is an existence statement and not a dimension count.

5. A Kadec–Avdonin criterion with blocks of variable length

5.1. **Discrepancy and generating function.**

Definition 5.1 (Discrepancy, Poisson and conjugate operators)**.** Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ be a real nondecreasing sequence with

$$|\mathrm{x}_k - (k - \tfrac{1}{2})| \le \mathrm{c}_{\mathscr{X}}\log(2+|k|) \qquad (k \in \mathbb{Z}). \tag{5.1}$$

Its *discrepancy* with respect to the half-integers is

$$\Psi_{\mathscr{X}}(\mathrm{x}) := \sum_{k\in\mathbb{Z}} \big(\mathbf{1}_{[\mathrm{x}_k,\infty)}(\mathrm{x}) - \mathbf{1}_{[k-\frac{1}{2},\infty)}(\mathrm{x})\big). \tag{5.2}$$

The k-th term is ± 1 on the interval between x_k and $k - \frac{1}{2}$ and 0 elsewhere. By (5.1), at most $O(\log(2+|\mathrm{x}|))$ terms are nonzero at x, so $|\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'_{\mathscr{X}}\log(2+|\mathrm{x}|)$ and $\Psi_{\mathscr{X}} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$. For $\mathrm{g} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$ and $\mathrm{y} > 0$ put

$$\mathbb{P}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \frac{\mathrm{y}\,\mathrm{g}(\mathrm{s})\,d\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}, \qquad \mathbb{Y}_{\mathrm{y}}\mathrm{g}(\mathrm{x}) := \frac{1}{\pi}\int \mathrm{g}(\mathrm{s})\Big[\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\Big]d\mathrm{s}. \tag{5.3}$$

The last term in $\mathbb{Y}_{\mathrm{y}}$ only makes the integral converge. Changing it changes $\mathbb{Y}_{\mathrm{y}}\mathrm{g}$ by a constant, which is irrelevant for all statements below.

Lemma 5.2 (Generating function)**.** *Let $\mathscr{X} = (\mathrm{x}_k)_{k\in\mathbb{Z}}$ satisfy* (5.1)*, understood as a sequence indexed by $\mathbb{Z}$ in which a point of multiplicity m occurs m times. Let $\mathcal{F}_0(\mathfrak{z}) := \cos \pi\mathfrak{z}$, and fix $\mathfrak{z}_0 \in \mathbb{R} \setminus (\mathscr{X} \cup (\mathbb{Z} + \frac{1}{2}))$. The series*

$$\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) := \sum_{k\in\mathbb{Z}} \Big(\frac{1}{\mathfrak{z} - \mathrm{x}_k} - \frac{1}{\mathfrak{z} - k + \frac{1}{2}} \Big) \tag{5.4}$$

converges absolutely and locally uniformly off $\mathscr{X} \cup (\mathbb{Z} + \frac{1}{2})$, and

$$\mathcal{F}_{\mathscr{X}}(\mathfrak{z}) := \mathcal{F}_0(\mathfrak{z}) \exp \int_{\mathfrak{z}_0}^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}(\mathrm{z})\, d\mathrm{z} \tag{5.5}$$

is a real entire function whose zero set is $\mathscr{X}$, with multiplicities. Moreover:

(i) *for $\mathrm{y} > 0$ there is a constant $\mathrm{c}''_{\mathscr{X}}$ such that*

$$\log|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y})| = \log|\mathcal{F}_0(\mathrm{x}+i\mathrm{y})| + \pi\, \mathbb{Y}_{\mathrm{y}} \Psi_{\mathscr{X}}(\mathrm{x}) + \mathrm{c}''_{\mathscr{X}}; \tag{5.6}$$

(ii) *$\mathcal{F}_{\mathscr{X}}$ is of exponential type π, and its indicator diagram is the segment $[-i\pi, i\pi]$;*

(iii) *if $\mathscr{X}$ is symmetric, $\mathrm{x}_{1-k} = -\mathrm{x}_k$, and $\mathfrak{z}_0 = 0 \notin \mathscr{X}$, then $\mathcal{F}_{\mathscr{X}}$ is even.*

Proof. Each term of $\mathfrak{G}_{\mathscr{X}}$ equals $\frac{x_k - k + \frac{1}{2}}{(\mathfrak{z}-\mathrm{x}_k)(\mathfrak{z}-k+\frac{1}{2})} = O(\log(2+|k|)/k^2)$ locally uniformly, so the series converges. $\mathfrak{G}_{\mathscr{X}}$ is meromorphic with simple poles, with residue $+m$ at a point of $\mathscr{X}$ of multiplicity m and -1 at each half-integer; coincident poles combine their residues. The residues are integers, so $\exp\int_{\mathfrak{z}_0}^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}$ is single-valued. It has a pole of order one at each half-integer not in $\mathscr{X}$, cancelled by the zero of $\mathcal{F}_0$. Hence $\mathcal{F}_{\mathscr{X}}$ is entire with zero set $\mathscr{X}$. $\mathfrak{G}_{\mathscr{X}}(\bar{\mathfrak{z}}) = \overline{\mathfrak{G}_{\mathscr{X}}(\mathfrak{z})}$ and $\mathfrak{z}_0$ is real, so $\mathcal{F}_{\mathscr{X}}$ is real on $\mathbb{R}$. Under the symmetry in (iii), $\mathfrak{G}_{\mathscr{X}}$ is odd, so $\int_0^{\mathfrak{z}} \mathfrak{G}_{\mathscr{X}}$ is even, and $\mathcal{F}_{\mathscr{X}}$ is even.

(i) For $\Im\mathfrak{z} \neq 0$ put $\mathbb{A}_{\mathscr{X}}(\mathfrak{z}) := \int \Psi_{\mathscr{X}}(\mathrm{s}) [\frac{1}{\mathfrak{z}-\mathrm{s}} + \frac{\mathrm{s}}{1+\mathrm{s}^2}] d\mathrm{s}$. For $f(\mathrm{s}) := \frac{1}{\mathfrak{z}-\mathrm{s}}$, the identity $f(\mathrm{x}_k) - f(k - \frac{1}{2}) = -\int f'(\mathrm{s})(\mathbf{1}_{[\mathrm{x}_k,\infty)} - \mathbf{1}_{[k-\frac{1}{2},\infty)})(\mathrm{s})\, d\mathrm{s}$ summed over k (Fubini applies because $\sum_k \int |f'||\cdots| < \infty$ by (5.1)) gives $\mathfrak{G}_{\mathscr{X}}(\mathfrak{z}) = -\int \Psi_{\mathscr{X}}(\mathrm{s})(\mathfrak{z}-\mathrm{s})^{-2} d\mathrm{s} = \mathbb{A}'_{\mathscr{X}}(\mathfrak{z})$. So $\mathcal{F}_{\mathscr{X}} = \mathrm{c}_+ \mathcal{F}_0 e^{\mathbb{A}_{\mathscr{X}}}$ on $\Im\mathfrak{z} > 0$, with $\mathrm{c}_+ \neq 0$. Since $\Re\frac{1}{\mathfrak{z}-\mathrm{s}} = \frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2}$, $\Re\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = \pi \mathbb{Y}_{\mathrm{y}} \Psi_{\mathscr{X}}(\mathrm{x})$ and $\Im\mathbb{A}_{\mathscr{X}}(\mathrm{x}+i\mathrm{y}) = -\pi \mathbb{P}_{\mathrm{y}} \Psi_{\mathscr{X}}(\mathrm{x})$.

(ii) Split $\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})$ into $|\mathrm{s}-\mathrm{x}| \le 1$, $1 < |\mathrm{s}-\mathrm{x}|$ with $|\mathrm{s}| \le 2|\mathfrak{z}| + 2$, and $|\mathrm{s}| > 2|\mathfrak{z}| + 2$. Using $|\Psi_{\mathscr{X}}(\mathrm{s})| \le \mathrm{c}'_{\mathscr{X}} \log(2+|\mathrm{s}|)$, $\int_{|u|\le 1} \frac{|u|\, du}{u^2+\mathrm{y}^2} = \log(1+\mathrm{y}^{-2})$, and, for $|\mathrm{s}| > 2|\mathfrak{z}| + 2$, $|\frac{\mathrm{x}-\mathrm{s}}{(\mathrm{x}-\mathrm{s})^2+\mathrm{y}^2} + \frac{\mathrm{s}}{1+\mathrm{s}^2}| \le 8(1+|\mathfrak{z}|)/\mathrm{s}^2$, one gets

$$|\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(\mathrm{x})| \le \mathrm{c}'''_{\mathscr{X}} (1 + \log(1+\mathrm{y}^{-2})) \log^2(4+|\mathfrak{z}|) \qquad (\mathfrak{z} = \mathrm{x}+i\mathrm{y},\ \mathrm{y} > 0). \tag{5.7}$$

For $|\mathrm{y}| \ge 1$, $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z})| \le \pi|\mathrm{y}| + O(\log^2(4+|\mathfrak{z}|))$ by (5.6), (5.7) and $|\mathcal{F}_{\mathscr{X}}(\bar{\mathfrak{z}})| = |\mathcal{F}_{\mathscr{X}}(\mathfrak{z})|$. For $|\mathrm{y}_1| < 1$, apply the sub-mean-value property of $\log|\mathcal{F}_{\mathscr{X}}|$ on the circle $|\mathfrak{z} - \mathfrak{z}_1| = \frac{1}{2}$. The factor $\log(1+\mathrm{y}^{-2})$ is integrable along the circle uniformly in y_1, since it has at most two logarithmic singularities. This gives $\log|\mathcal{F}_{\mathscr{X}}(\mathfrak{z}_1)| \le \frac{3\pi}{2} + O(\log^2(4+|\mathfrak{z}_1|))$ for $|\mathrm{y}_1| < 1$. Hence the type is at most π and the indicator satisfies $\mathrm{h}(\mathrm{u}) \le \pi|\sin\mathrm{u}|$. Since (5.6) is an equality and (5.7) is two-sided, $\log|\mathcal{F}_{\mathscr{X}}(i\mathrm{y})| = \log|\cos(i\pi\mathrm{y})| + \pi\mathbb{Y}_{\mathrm{y}}\Psi_{\mathscr{X}}(0) + \mathrm{c}''_{\mathscr{X}} = \pi|\mathrm{y}| + O(\log^2(4+|\mathrm{y}|))$ for $|\mathrm{y}| \ge 1$, so $\mathrm{h}(\pm\frac{\pi}{2}) = \pi$. The indicator diagram lies in $[-i\pi, i\pi]$, whose support function is $\pi|\sin\mathrm{u}|$, and it contains $\pm i\pi$, so it equals that segment. □

5.2. **The criterion.** A weight $\mathrm{w} > 0$ on $\mathbb{R}$ is in (A_2) if

$$[\mathrm{w}]_{A_2} := \sup_I \Big(\frac{1}{|I|} \int_I w \Big) \Big(\frac{1}{|I|} \int_I w^{-1} \Big) < \infty, \tag{5.8}$$

the supremum being over bounded intervals.

Theorem 5.3 (Block $\frac{1}{4}$-criterion)**.** *Let $\mathscr{X}$ satisfy* (5.1) *and suppose $\Psi_{\mathscr{X}} = \mathrm{A} + \mathrm{B}$ with measurable A, B such that:*

(i) $\|\mathrm{A}\|_\infty \le \mathrm{d}_\star < \frac{1}{4}$;

(ii) $\sup_x |\mathbb{Y}_1\mathrm{B}(\mathrm{x}) - \mathrm{c_B}| \le \mathbb{U}$ *for some constant* $\mathrm{c_B}$.

Then $\mathrm{w}(\mathrm{x}) := |\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2$ *is in* (A_2) *and*

$$[\mathrm{w}]_{A_2} \le \frac{\pi^2 \coth^2(\pi)\, e^{4\pi\mathbb{U}}}{\cos^2(2\pi \mathrm{d}_\star)}. \tag{5.9}$$

Proof. $\mathrm{B} = \Psi_{\mathscr{X}} - \mathrm{A} \in L^1(d\mathrm{x}/(1+\mathrm{x}^2))$, so $\mathbb{Y}_1\Psi_{\mathscr{X}} = \mathbb{Y}_1\mathrm{A} + \mathbb{Y}_1\mathrm{B}$. By (5.6), $\mathrm{w} = e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{A}} e^{2\pi\mathbb{Y}_1\mathrm{B}}$.

Put $\mathbb{A}_\mathrm{A}(\mathfrak{z}) := \int \mathrm{A}(\mathrm{s})\big[\frac{1}{\mathfrak{z}-\mathrm{s}} + \frac{\mathrm{s}}{1+\mathrm{s}^2}\big]d\mathrm{s}$. It is holomorphic on $\Im\mathfrak{z} > 0$, and $\mathbb{A}_\mathrm{A}(\mathfrak{o}+i)$ is holomorphic on $\Im\mathfrak{o} > -1$. Let $\mathcal{W}(\mathfrak{o}) := \exp(2\mathbb{A}_\mathrm{A}(\mathfrak{o}+i))$. For $\mathfrak{o} = \mathrm{x}_1 + i\mathrm{y}_1$ with $\mathrm{y}_1 \ge 0$,

$$|\mathcal{W}(\mathfrak{o})| = e^{2\pi\mathbb{Y}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1)}, \qquad \arg\mathcal{W}(\mathfrak{o}) = -2\pi\mathbb{P}_{1+\mathrm{y}_1}\mathrm{A}(\mathrm{x}_1) \in [-2\pi\mathrm{d}_\star, 2\pi\mathrm{d}_\star]. \tag{5.10}$$

So $\mathcal{W}$ maps the closed upper half-plane into the sector $|\arg| \le \mathrm{b}_\star := 2\pi\mathrm{d}_\star < \frac{\pi}{2}$, and $|\mathcal{W}(\mathrm{x})| = e^{2\pi\mathbb{Y}_1\mathrm{A}(\mathrm{x})}$ on $\mathbb{R}$.

$\Re\mathcal{W}$ is a positive harmonic function on $\Im\mathfrak{o} > 0$, continuous up to $\mathbb{R}$. By the Herglotz representation it equals $\mathrm{c}\,\mathrm{y}_1 + \int \mathbb{P}_\mathfrak{o}(t)\,d\mathrm{n}(t)$ with $\mathrm{c} \ge 0$, $\mathrm{n} \ge 0$, and $\mathrm{n} = \Re\mathcal{W}(t)\,dt$ by continuity. Here $\mathbb{P}_\mathfrak{o}(t) := \frac{1}{\pi}\frac{\mathrm{y}_1}{(t-\mathrm{x}_1)^2+\mathrm{y}_1^2}$. Hence $\Re\mathcal{W}(\mathfrak{o}) \ge \int \mathbb{P}_\mathfrak{o}(t)\Re\mathcal{W}(t)\,dt$. For an interval I with centre x_I, put $\mathfrak{o}_I := \mathrm{x}_I + i|I|/2$. Then $\mathbb{P}_{\mathfrak{o}_I}(t) \ge (\pi|I|)^{-1}$ for $t \in I$, and

$$\frac{1}{|I|}\int_I |\mathcal{W}| \le \frac{1}{\cos\mathrm{b}_\star}\frac{1}{|I|}\int_I \Re\mathcal{W} \le \frac{\pi}{\cos\mathrm{b}_\star}\Re\mathcal{W}(\mathfrak{o}_I) \le \frac{\pi}{\cos\mathrm{b}_\star}|\mathcal{W}(\mathfrak{o}_I)|. \tag{5.11}$$

$1/\mathcal{W}$ maps into the same sector, so the same holds for it. Multiplying, $[|\mathcal{W}|]_{A_2} \le \pi^2/\cos^2\mathrm{b}_\star$.

Finally $\mathrm{w} = \mathrm{g}\,|\mathcal{W}|$ with $\mathrm{g} := e^{2\mathrm{c}''_{\mathscr{X}}}|\mathcal{F}_0(\mathrm{x}+i)|^2 e^{2\pi\mathbb{Y}_1\mathrm{B}}$. Since $|\cos\pi(\mathrm{x}+i)|^2 = \cos^2(\pi\mathrm{x})\cosh^2\pi + \sin^2(\pi\mathrm{x})\sinh^2\pi \in [\sinh^2\pi, \cosh^2\pi]$ and $e^{2\pi\mathbb{Y}_1\mathrm{B}} \in [e^{2\pi(\mathrm{c_B}-\mathbb{U})}, e^{2\pi(\mathrm{c_B}+\mathbb{U})}]$, we have $\sup\mathrm{g}/\inf\mathrm{g} \le \coth^2(\pi)e^{4\pi\mathbb{U}}$. For any weight w_1, $[\mathrm{g}\mathrm{w}_1]_{A_2} \le (\sup\mathrm{g}/\inf\mathrm{g})[\mathrm{w}_1]_{A_2}$. □

Corollary 5.4 (Riesz bases)**.** *Under the hypotheses of Theorem 5.3:*

(a) *if* $\mathscr{X}$ *is uniformly discrete (in particular without repeated points),* $\{e^{i\mathrm{x}_k\mathrm{v}}\}_{k\in\mathbb{Z}}$ *is a Riesz basis of* $L^2(-\pi,\pi)$*;*

(b) *if* $\mathscr{X}$ *is a finite union of uniformly discrete sets, the generalised divided differences of* $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ *in the sense of* [3] *form a Riesz basis of* $L^2(-\pi,\pi)$*. Consequently, for every partition of* $\mathbb{Z}$ *into consecutive blocks* $\mathfrak{B}_m$ *that are unions of the groups of* [3]*, the subspaces spanned by* $\{\mathrm{v}^j e^{i\mathrm{x}\mathrm{v}}\}$*, with* $\mathrm{x} = \mathrm{x}_k$ *for* $k \in \mathfrak{B}_m$ *and* j *below the multiplicity of* x*, form a Riesz basis of subspaces.*

Proof. By Lemma 5.2, $\mathcal{F}_{\mathscr{X}}$ is entire of exponential type with indicator diagram of width 2π and zero set $\mathscr{X}$. By Theorem 5.3, $|\mathcal{F}_{\mathscr{X}}(\mathrm{x}+i)|^2 \in (A_2)$, which is equivalent to the Helson–Szegő condition [9, Ch. VI]. Then (a) is Pavlov's criterion [3, Thm. 1], and the first statement of (b) is [3, Thm. 3]. Both are stated for $L^2(0,2\pi)$, and translation by π multiplies each exponential by a unimodular constant. Grouping the vectors of a Riesz basis into disjoint groups gives a Riesz basis of subspaces. □

5.3. **Kadec, Avdonin, and variable block length.** Write $\mathrm{x}_k = k - \frac{1}{2} + \mathrm{d}_k$, $I_k := [k-1, k)$, $\mathsf{i}_k := \mathbf{1}_{[\mathrm{x}_k,\infty)} - \mathbf{1}_{[k-\frac{1}{2},\infty)}$, so that $\Psi_{\mathscr{X}} = \sum_k \mathsf{i}_k$ and $\int \mathsf{i}_k = -\mathrm{d}_k$.

Proposition 5.5 (Bounded conjugate for block-supported oscillation)**.** *Let* $\{J_m\}$ *be a partition of* $\mathbb{R}$ *into consecutive intervals of lengths* $\mathfrak{n}_m \ge 1$*. Let* $\mathrm{B} = \sum_m B_m$ *with* $\operatorname{supp}\mathrm{B}_m \subset \mathfrak{J}_m$*,* $\int \mathrm{B}_m = 0$ *and* $|\mathrm{B}_m| \le \mathrm{D}'_m$*,* $\sup_m \mathrm{D}'_m < \infty$*. Put*

$$\mathbb{U}_{\mathrm{loc}} := \sup_m \sup_{\mathrm{x}\in\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}} \Big|\int_{J_m} \frac{(\mathrm{x}-\mathrm{s})\,\mathrm{B}_m(\mathrm{s})}{\pi((\mathrm{x}-\mathrm{s})^2+1)}\,d\mathrm{s}\Big|, \qquad \mathbb{O} := \sup_{m_0} \sum_{|m-m_0|\ge 2} \frac{N_m^2}{\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)^2}. \tag{5.12}$$

If $\mathbb{U}_{\mathrm{loc}} < \infty$ *and* $\mathbb{O} < \infty$*, then* $\sup_x |\mathbb{Y}_1\mathrm{B}(\mathrm{x}) - \mathrm{c_B}| \le 3\mathbb{U}_{\mathrm{loc}} + \frac{\mathbb{O}}{2\pi}\sup_m \mathrm{D}'_m$ *for a constant* $\mathrm{c_B}$.

Proof. The term $\frac{1}{\pi}\int \frac{\mathrm{s}}{1+\mathrm{s}^2}\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$ equals $\frac{1}{\pi}\int(\frac{\mathrm{s}}{1+\mathrm{s}^2}-\frac{\mathrm{m}_m}{1+\mathrm{m}_m^2})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}$, where m_m is the centre of $\mathfrak{J}_m$. Since $|\frac{d}{d\mathrm{s}}\frac{\mathrm{s}}{1+\mathrm{s}^2}|\le\frac{1}{1+\mathrm{s}^2}$, its modulus is at most $\mathrm{D}'_m N_m^2/(2\pi(1+\operatorname{dist}(0,\mathfrak{J}_m)^2))$. If m_0 is the index with $0\in\mathfrak{J}_{m_0}$, then $\operatorname{dist}(0,\mathfrak{J}_m)\ge\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$, so these terms are summable by $\mathbb{O}<\infty$. Their sum is the constant $\mathrm{c_B}$. Let $\mathrm{x}\in\mathfrak{J}_{m_0}$. The three blocks with $|m-m_0|\le 1$ contribute at most $3\mathbb{U}_{\mathrm{loc}}$. For $|m-m_0|\ge 2$, $\int \mathrm{B}_m=0$ gives $\big|\int_{J_m}\mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_m(\mathrm{s})\,d\mathrm{s}\big|\le \mathrm{D}'_m N_m\cdot\frac{N_m}{2}\sup_{\mathrm{s}\in\mathfrak{J}_m}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|$ with $\mathrm{k}(\mathrm{v}):=\frac{\mathrm{v}}{\pi(\mathrm{v}^2+1)}$. Moreover $|\mathrm{k}'(\mathrm{v})|\le\frac{1}{\pi\mathrm{v}^2}$ and $|\mathrm{x}-\mathrm{s}|\ge\operatorname{dist}(\mathfrak{J}_{m_0},\mathfrak{J}_m)$. Sum over m. □

Corollary 5.6 (Kadec and Avdonin, with the constant $\frac{1}{4}$)**.** (a) [11] *If $|\mathrm{d}_k|\le\mathrm{d}_\star<\frac{1}{4}$ for all k, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*
(b) [2] *If $\mathscr{X}$ is uniformly discrete, $\sup_k|\mathrm{d}_k|\le\mathrm{D}$, and for a fixed $\mathfrak{n}$ the means of d_k over consecutive blocks of $\mathfrak{n}$ indices are at most $\mathrm{d}_\star<\frac{1}{4}$ in modulus, then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$.*

Proof. Both statements follow from Theorem 5.7 below: in (a) take $\mathfrak{n}_m\equiv 1$, and in (b) take $\mathfrak{n}_m\equiv\mathfrak{n}$. In both cases the blocks at index distance $j\ge 2$ from $\mathfrak{J}_{m_0}$ are at distance $(j-1)\mathfrak{n}$, so $\mathbb{O}\le 2\sum_{j\ge2}\mathfrak{n}^2/((j-1)\mathfrak{n})^2=\pi^2/3$ and $\mathbb{V}\le 2\mathrm{D}\log(2+3\mathfrak{n})$; for $\mathfrak{n}=1$, $\mathrm{D}_m=0$ and $\mathbb{V}=0$. For (a), $\mathrm{d}_\star<\frac{1}{4}$ gives gaps $\ge 1-2L>\frac{1}{2}$, so $\mathscr{X}$ is uniformly discrete. □

Theorem 5.7 (Avdonin's theorem with variable block length)**.** *Let $\mathscr{X}$ be uniformly discrete with $\sup_k|\mathrm{d}_k|\le\mathrm{D}$. Let $\{\mathfrak{B}_m\}$ be a partition of $\mathbb{Z}$ into consecutive blocks of $\mathfrak{n}_m$ indices, and $\mathfrak{J}_m:=\bigcup_{k\in\mathfrak{B}_m}I_k$. Let $\bar{\mathrm{d}}_m$ be the mean of d_k over $\mathfrak{B}_m$, $\mathrm{D}_m:=\max_{k\in\mathfrak{B}_m}|\mathrm{d}_k-\bar{\mathrm{d}}_m|$, and $\mathfrak{n}_m^*:=\mathfrak{n}_{m-1}+\mathfrak{n}_m+\mathfrak{n}_{m+1}$. Assume*

(a) $\sup_m|\bar{\mathrm{d}}_m|\le\mathrm{d}_\star<\frac{1}{4}$;
(b) $\mathbb{O}<\infty$, *with $\mathbb{O}$ as in* (5.12);
(c) $\mathbb{V}:=\sup_m \mathrm{D}_m\log(2+\mathfrak{n}_m^*)<\infty$.

Then $\{e^{i\mathrm{x}_k\mathrm{v}}\}$ is a Riesz basis of $L^2(-\pi,\pi)$, and (5.9) *holds with*

$$\mathbb{U}=16(1+\mathrm{D})^3+\frac{6\mathbb{V}}{\pi}+\frac{V\mathbb{O}}{2\pi\log 2}. \tag{5.13}$$

Proof. Write $\Psi_{\mathscr{X}}=\mathrm{A}+\mathrm{B}_1+\mathrm{B}_2$ with $\mathrm{A}:=-\sum_m\bar{\mathrm{d}}_m\mathbf{1}_{J_m}$, $\mathrm{B}_1:=\sum_k b_k$ where $\mathrm{b}_k:=\mathrm{i}_k+\mathrm{d}_k\mathbf{1}_{I_k}$, and $\mathrm{B}_2:=-\sum_m\sum_{k\in\mathfrak{B}_m}(\mathrm{d}_k-\bar{\mathrm{d}}_m)\mathbf{1}_{I_k}$. Then $\|\mathrm{A}\|_\infty\le d$.

B_1. b_k has mean zero, $|\mathrm{b}_k|\le 1+\mathrm{D}$, and support in an interval of length at most $2\mathrm{D}+1$ containing $\mathrm{m}_k:=k-\frac{1}{2}$. With k as above, $\sup|\mathrm{k}|=\frac{1}{2\pi}$ and $|\mathrm{k}'(\mathrm{v})|\le\frac{1}{\pi(1+\mathrm{v}^2)}$, so $|\int\mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}|=|\int(\mathrm{k}(\mathrm{x}-\mathrm{s})-\mathrm{k}(\mathrm{x}-\mathrm{m}_k))\mathrm{b}_k(\mathrm{s})\,d\mathrm{s}|\le(1+\mathrm{D})(2\mathrm{D}+1)\min\big(\frac{1}{\pi},(2\mathrm{D}+1)\sup_{\mathrm{s}\in\operatorname{supp}\mathrm{b}_k}|\mathrm{k}'(\mathrm{x}-\mathrm{s})|\big)$. At most $8\mathrm{D}+5\le 8(1+\mathrm{D})$ indices have $|\mathrm{x}-\mathrm{m}_k|<2(2\mathrm{D}+1)$, and they contribute at most $16(1+\mathrm{D})^3/\pi$. For the others $|\mathrm{x}-\mathrm{s}|\ge|\mathrm{x}-\mathrm{m}_k|/2$ on the support, so the term is at most $4(1+\mathrm{D})^3\cdot\frac{4}{\pi}(4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1}$. Since $\sum_k(4+(\mathrm{x}-\mathrm{m}_k)^2)^{-1}\le\frac{1}{2}+\frac{\pi}{2}$, they contribute at most $\frac{16}{\pi}(\frac{1}{2}+\frac{\pi}{2})(1+\mathrm{D})^3$. The total is below $16(1+\mathrm{D})^3$. The modifying term of $\mathbb{Y}_1$ contributes a convergent sum of constants, as in Proposition 5.5.

B_2. On $\mathfrak{J}_m$ it has mean zero and modulus at most D_m. For x in $\mathfrak{J}_{m-1}\cup\mathfrak{J}_m\cup\mathfrak{J}_{m+1}$, $|\int_{J_m}\mathrm{k}(\mathrm{x}-\mathrm{s})\mathrm{B}_2(\mathrm{s})\,d\mathrm{s}|\le\frac{\mathrm{D}_m}{\pi}\int_{|u|\le\mathfrak{n}_m^*}\frac{|u|\,du}{1+u^2}\le\frac{2}{\pi}\mathrm{D}_m\log(2+\mathfrak{n}_m^*)\le\frac{2\mathbb{V}}{\pi}$. Moreover $\sup_m\mathrm{D}_m\le\mathbb{V}/\log 2$. Proposition 5.5 gives $|\mathbb{Y}_1\mathrm{B}_2-\mathrm{c}|\le\frac{6\mathbb{V}}{\pi}+\frac{V\mathbb{O}}{2\pi\log 2}$.

Apply Theorem 5.3 and Corollary 5.4(a). □

Remark 5.8 (The price of variable length). For fixed $\mathfrak{n}$, conditions (b) and (c) hold automatically, and Theorem 5.7 is Avdonin's theorem. If $\mathfrak{n}_m\to\infty$, the averaging condition (a) is still only $|\bar{\mathrm{d}}_m|\le\mathrm{d}_\star<\frac{1}{4}$. However, (c) then requires the oscillation inside the block to decay like $1/\log\mathfrak{n}_m$. In Theorem 5.3 the oscillation enters only through the bounded-conjugate condition (ii), which allows cancellation inside the blocks and is weaker than (c). Without some condition of this kind the conclusion fails for growing blocks: a block of length $\mathfrak{n}$ with oscillation of size D_m can

make $|\mathbb{Y}_1\mathrm{B}|$ of size $\mathrm{D}_m \log \mathfrak{n}$. Lyubarskii and Seip derive Kadec's theorem from their Muckenhoupt characterisation ([13, Thm. 2] for $p = 2$), and indicate that the same technique gives results of Avdonin type. I have not searched the literature for Theorem 5.7 or for the bounded-conjugate formulation of Theorem 5.3, so their novelty is not claimed. The proof of Theorem 5.3 is the sufficiency half of the Helson–Szegő argument, applied on the line $\Im \mathfrak{z} = 1$.

6. Application to lists near the RvM clock

6.1. Unfolding.

Definition 6.1 (Unfolded list)**.** Let $\mathfrak{t}_0 := 9.6769\ldots$ be the zero of $\mathcal{N}_{\mathrm{RVM}}$ in $(2\pi, \infty)$. For a C-admissible list $\boldsymbol{\kappa}$ with $\kappa_1 > \mathfrak{t}_0$ (true for $\boldsymbol{\tau}$ and $\boldsymbol{\gamma}$), put $\mathrm{x}_k := \mathcal{N}_{\mathrm{RVM}}(\kappa_k) > 0$ for $k \geq 1$ and $\mathrm{x}_{1-k} := -\mathrm{x}_k$. The resulting symmetric sequence $\mathscr{X}_{\boldsymbol{\kappa}}$ has discrepancy $\Psi_{\boldsymbol{\kappa}} := \Psi_{\mathscr{X}_{\boldsymbol{\kappa}}}$, and $\mathcal{F}_{\boldsymbol{\kappa}} := F_{\mathscr{X}_{\boldsymbol{\kappa}}}$ with $\mathfrak{z}_0 = 0$. For the clock, $\mathscr{X}_{\boldsymbol{\tau}} = \mathbb{Z} + \frac{1}{2}$ and $\mathcal{F}_{\boldsymbol{\tau}} = \mathcal{F}_0$.

Lemma 6.2 (Unfolded discrepancy)**.** *For a C-admissible list with $\kappa_1 > \mathfrak{t}_0$, $|\mathrm{x}_k - (k - \frac{1}{2})| \leq \frac{C}{2\pi} \log \frac{\tau_k + C}{2\pi}$ for $k \geq 1$, so (5.1) holds. Moreover, for $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$ with $\mathrm{x} > 0$,*

$$\Psi_{\boldsymbol{\kappa}}(\mathrm{x}) = N_{\boldsymbol{\kappa}}(t) - n_{\mathrm{RVM}}(t), \qquad |N_{\boldsymbol{\kappa}}(t) - n_{\mathrm{RVM}}(t)| \leq \frac{C}{\pi} \log \frac{t + C}{2\pi} + 2, \tag{6.1}$$

and $\Psi_{\boldsymbol{\kappa}}(-\mathrm{x}) = -\Psi_{\boldsymbol{\kappa}}(\mathrm{x})$ for almost every x.

Proof. By the mean value theorem and the monotonicity of $\mathcal{N}'_{\mathrm{RVM}}$, $|\mathcal{N}_{\mathrm{RVM}}(\kappa_k) - \mathcal{N}_{\mathrm{RVM}}(\tau_k)| \leq C\mathcal{N}'_{\mathrm{RVM}}(\tau_k + C)$. For $\mathrm{x} > 0$ the terms of (5.2) with $k \leq 0$ vanish, because both x_k and $k - \frac{1}{2}$ are negative. For $k \geq 1$, $\mathrm{x}_k \leq \mathrm{x}$ if and only if $\kappa_k \leq t$, by monotonicity of $\mathcal{N}_{\mathrm{RVM}}$. If $N_{\boldsymbol{\kappa}}(t) = j$, then $\tau_j \leq t + C$ and $\tau_{j+1} > t - C$, so $n_{\mathrm{RVM}}(t - C) \leq j \leq n_{\mathrm{RVM}}(t + C)$. By (2.2), $n_{\mathrm{RVM}}(t + C) - n_{\mathrm{RVM}}(\max(t - C, 2\pi)) \leq \mathcal{N}_{\mathrm{RVM}}(t + C) - \mathcal{N}_{\mathrm{RVM}}(\max(t - C, 2\pi)) + 1$. This is at most $\frac{2C}{2\pi} \log \frac{t+C}{2\pi} + 2$, the extra 1 covering $t - C < 2\pi$, where $\mathcal{N}_{\mathrm{RVM}} \geq -\frac{1}{8}$ and $n_{\mathrm{RVM}} = 0$. Oddness follows from the symmetry of $\mathscr{X}_{\boldsymbol{\kappa}}$ and of $\mathbb{Z} + \frac{1}{2}$. □

Proposition 6.3 (The condition of $\mathscr{O}$ gives the averaging hypothesis with blocks $\asymp \log^2 T$)**.** *Let $\boldsymbol{\kappa}$ be C-admissible with $\kappa_1 > \mathfrak{t}_0$, and assume the third condition of $\mathscr{O}$ (`eq: list-space`) in the form*

$$\Big| \int_{2\pi}^{R} (N_{\boldsymbol{\kappa}}(t) - \mathcal{N}_{\mathrm{RVM}}(t))\, dt \Big| \leq C_2 \log R \qquad (R \geq 3). \tag{6.2}$$

Put $\mathrm{H}(T) := 16 C_2 \log T$. There is $T_1 = T_1(C, C_2)$ such that the following holds for the points $T_{m+1} := T_m + \mathrm{H}(T_m)$ ($m \geq 1$) and the blocks $\mathfrak{J}_m := [\mathcal{N}_{\mathrm{RVM}}(T_m), \mathcal{N}_{\mathrm{RVM}}(T_{m+1}))$ with their mirror images $-\mathfrak{J}_m$. The step function $\mathrm{A}_{\boldsymbol{\kappa}}$, equal to the mean of $\Psi_{\boldsymbol{\kappa}}$ on each $\pm\mathfrak{J}_m$ and to 0 on $(-\mathcal{N}_{\mathrm{RVM}}(T_1), \mathcal{N}_{\mathrm{RVM}}(T_1))$, satisfies

$$\|\mathrm{A}_{\boldsymbol{\kappa}}\|_\infty \leq \tfrac{3}{16}, \qquad |\mathfrak{J}_m| = \frac{8 C_2}{\pi} \log^2 T_m \,(1 + o(1)) \quad (m \to \infty). \tag{6.3}$$

The remainder $\mathrm{B}_{\boldsymbol{\kappa}} := \Psi_{\boldsymbol{\kappa}} - \mathrm{A}_{\boldsymbol{\kappa}}$ has mean zero on every block. For $\boldsymbol{\kappa} = \boldsymbol{\gamma}$, (6.2) holds unconditionally for some C_2.

Proof. Write $\mathfrak{J} = [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(T + \mathrm{H}))$. Then

$$\begin{aligned} \int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\, d\mathrm{x} &= \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(t)\, \mathcal{N}'_{\mathrm{RVM}}(t)\, dt \\ &= \mathcal{N}'_{\mathrm{RVM}}(T) \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) + \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(\mathcal{N}'_{\mathrm{RVM}}(t) - \mathcal{N}'_{\mathrm{RVM}}(T))\, dt. \end{aligned} \tag{6.4}$$

The last integral is at most $\mathrm{H} \cdot \frac{\mathrm{H}}{2\pi T} (\frac{C}{\pi} \log \frac{T + \mathrm{H} + C}{2\pi} + 1)$, by (6.1) and $\mathcal{N}''_{\mathrm{RVM}}(t) = \frac{1}{2\pi t}$.

Next, $\int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}) = \int_T^{T+\mathrm{H}} (N_{\boldsymbol{\kappa}} - \mathcal{N}_{\mathrm{RVM}}) + \int_T^{T+\mathrm{H}} (\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})$. The first piece is at most $2C_2 \log(T + \mathrm{H})$ by (6.2). For the second, substitute $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$. By (2.2), $\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}} =$

$\mathrm{saw}(\mathrm{x}) := \mathrm{x} - \lfloor \mathrm{x} + \frac{1}{2} \rfloor$, which is 1-periodic with mean zero and $|\mathrm{saw}| \le \frac{1}{2}$, and $dt = g(\mathrm{x})\,d\mathrm{x}$ with $g := 1/\mathcal{N}'_{\mathrm{RVM}}(t(\mathrm{x}))$ positive and decreasing. On a full period $[j - \frac{1}{2}, j + \frac{1}{2})$, $|\int \mathrm{saw}\,\mathfrak{g}_1| = |\int \mathrm{saw}\,(\mathfrak{g}_1 - \mathfrak{g}_1(j))| \le \frac{1}{4}(\mathfrak{g}_1(j - \frac{1}{2}) - \mathfrak{g}_1(j + \frac{1}{2}))$, which telescopes. The two partial periods contribute at most $\mathfrak{g}_1(\mathcal{N}_{\mathrm{RVM}}(T))$. Hence $|\int_T^{T+\mathrm{H}} (\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}})| \le \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)}$.

Since $|\mathfrak{J}| \ge \mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}$,

$$\Big|\frac{1}{|\mathfrak{J}|}\int_{\mathfrak{J}} \Psi_{\boldsymbol{\kappa}}\Big| \le \frac{2C_2 \log(T+\mathrm{H})}{\mathrm{H}} + \frac{5}{4\mathcal{N}'_{\mathrm{RVM}}(T)\mathrm{H}} + \frac{\mathrm{H}}{2\pi T \mathcal{N}'_{\mathrm{RVM}}(T)}\Big(\frac{C}{\pi}\log\frac{T+\mathrm{H}+C}{2\pi} + 1\Big). \tag{6.5}$$

With $\mathrm{H} = 16C_2 \log T$, the first term tends to $\frac{1}{8}$ and the others tend to 0, so the bound is at most $\frac{3}{16}$ for $T \ge T_1(C, C_2)$. The length is $|\mathfrak{J}_m| = \mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})\mathrm{H}(T_m)$ with $\mathrm{s} \in [T_m, T_{m+1}]$. The mirror blocks follow by oddness. For $\boldsymbol{\gamma}$: $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$ [17, §9.3], $\int_0^R S = O(\log R)$ [17, Thm. 9.9], and $1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}} = O(1/t)$. Finally, $\boldsymbol{\gamma}$ is C_η-admissible, unconditionally, by `lem:bounded-displacement`. In particular (5.1) holds for $\mathscr{X}_{\boldsymbol{\gamma}}$, by Lemma 6.2. □

Remark 6.4 (Correction of an earlier formulation). Proposition 6.3 is an implication. The averaging condition on blocks of length $\asymp \log^2 T$ does not imply (6.2): summing $|\int_{\mathfrak{J}_m} \Psi| \le \frac{3}{16}|\mathfrak{J}_m|$ over the blocks up to height R gives only $O(R \log R)$, not $O(\log R)$. The two conditions do not coincide.

Corollary 6.5 (Reduction to one condition). *For every $\boldsymbol{\kappa}$ as in Proposition 6.3: if $\mathbb{Y}_1\mathrm{B}_{\boldsymbol{\kappa}}$ is bounded up to a constant, then $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x}+i)|^2 \in (A_2)$, with constant* (5.9) *for $\mathrm{d}_\star = \frac{3}{16}$. Corollary 5.4(a) then holds for $\mathscr{X}_{\boldsymbol{\kappa}}$ if it is uniformly discrete, and Corollary 5.4(b) if it is a finite union of uniformly discrete sets.*

Proof. Lemma 6.2, Proposition 6.3 and Theorem 5.3. □

6.2. **The weight is a quotient of synthetic zetas.**

Proposition 6.6 (Weight identity). *Let $\boldsymbol{\kappa}$ be C-admissible and satisfy* (6.2). *For $0 < \hbar \le 1$ and $t \ge 2\pi$,*

$$\log\Big|\frac{\zeta_{\boldsymbol{\kappa}}}{\zeta_{\mathrm{RVM}}}\Big(\tfrac{1}{2} + \hbar + it\Big)\Big| = \int_{2\pi}^{\infty} \frac{(t-\mathrm{s})\,(N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}})(\mathrm{s})}{(t-\mathrm{s})^2 + \hbar^2}\,d\mathrm{s} + \mathrm{c}_{\boldsymbol{\kappa}} + \mathrm{m}_{\boldsymbol{\kappa},\hbar}(t), \tag{6.6}$$

where the integral converges as an improper integral and $\mathrm{c}_{\boldsymbol{\kappa}}$ is a constant. The remainder satisfies $|\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t)| \le \mathrm{c}(C, C_2)\log(2+t)/(1+t)$.

Proof. For $s = \frac{1}{2} + \hbar + it$, $|\frac{1}{4} + \kappa^2 + q(s)| = |\kappa^2 + (\hbar + it)^2| = |\kappa - t + i\hbar|\,|\kappa + t - i\hbar|$. Hence, with $\mathrm{l}_\hbar(u) := \frac{1}{2}\log(u^2 + \hbar^2)$,

$$\log\Big|\frac{P_{\boldsymbol{\kappa}}}{P_{\boldsymbol{\tau}}}(s)\Big| = \sum_k [\mathrm{l}_\hbar(t - \kappa_k) - \mathrm{l}_\hbar(t - \tau_k)] + \sum_k [\mathrm{l}_\hbar(t + \kappa_k) - \mathrm{l}_\hbar(t + \tau_k)] - \sum_k \log\frac{\frac{1}{4} + \kappa_k^2}{\frac{1}{4} + \tau_k^2}. \tag{6.7}$$

Put $\mathfrak{g}_1 := N_{\boldsymbol{\kappa}} - n_{\mathrm{RVM}}$ and $\mathfrak{G}_1(\mathrm{s}) := \int_{2\pi}^{\mathrm{s}} \mathfrak{g}_1$. By (6.2), by the sawtooth estimate in the proof of Proposition 6.3 on $[\tau_1, \infty)$, and by $|\mathcal{N}_{\mathrm{RVM}} - n_{\mathrm{RVM}}| \le \frac{1}{2}$ on $[2\pi, \tau_1]$, $|\mathfrak{G}_1(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$. For f smooth with $\mathrm{f}'(\mathrm{s}) = O(1/\mathrm{s})$ and $\mathrm{f}''(\mathrm{s}) = O(1/\mathrm{s}^2)$, the sums $\sum_{k \le k^*}[\mathrm{f}(\kappa_k) - \mathrm{f}(\tau_k)] = -\int \mathrm{f}'\mathfrak{g}_{1,k^*}$ converge as $k^* \to \infty$ to $-\int^\infty \mathrm{f}'\mathfrak{g}_1$, an improper integral. Integration by parts against $\mathfrak{G}_1$ shows both the convergence and that the tail contributions vanish. With $\mathrm{f} = \mathrm{l}_\hbar(t - \cdot)$ this gives the integral in (6.6). With $\mathrm{f} = \log(\frac{1}{4} + (\cdot)^2)$ it gives the constant $\mathrm{c}_{\boldsymbol{\kappa}}$. With $\mathrm{f} = \mathrm{l}_\hbar(t + \cdot)$ it gives $\mathrm{m}_{\boldsymbol{\kappa},\hbar}(t) = \int_{2\pi}^\infty \mathrm{k}_1'(\mathrm{s})\mathfrak{G}_1(\mathrm{s})\,d\mathrm{s}$, where $\mathrm{k}_1(\mathrm{s}) := \frac{t+\mathrm{s}}{(t+\mathrm{s})^2+\hbar^2}$ and $|\mathrm{k}_1'(\mathrm{s})| \le (t+\mathrm{s})^{-2}$. This yields the bound. □

Remark 6.7 (Local scale). With $\hbar = 1$ and $\hbar_T := \hbar/\mathcal{N}'_{\mathrm{RVM}}(T) = 2\pi/\log(T/2\pi)$ (the notation of the proof of Theorem 6.8), the substitution $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t)$ maps the kernel $\frac{(t-\mathrm{s})\,d\mathrm{s}}{(t-\mathrm{s})^2+\hbar_T^2}$ to $\frac{(\mathrm{x}-\mathrm{x}')\,d\mathrm{x}'}{(\mathrm{x}-\mathrm{x}')^2+1}$ to first order near $t = T$. Comparing (6.6) with (5.6) at $\mathrm{y} = 1$, the unfolded weight $|\mathcal{F}_{\boldsymbol{\kappa}}(\mathrm{x}+i)|^2$ near

$\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(T)$ is therefore, to first order, $|\zeta_{\boldsymbol{\kappa}}/\zeta_{\mathrm{RVM}}|^2$ on the line $\Re s = \frac{1}{2} + \hbar_T$, up to the bounded factor $|\mathcal{F}_0(\mathrm{x}+i)|^2$. The identity (6.6) is exact. The identification with the unfolded weight is local, and its quantitative form for the Poisson part is Step 3 of the proof of Theorem 6.8.

6.3. **Obstruction for the zeta list.**

Theorem 6.8 (No (A_2) for the unfolded zeta ordinates)**.** *Let $\mathscr{X}_{\boldsymbol{\gamma}}$ be the unfolded list of the ordinates $\gamma_k > 0$ of the nontrivial zeros of ζ, counted with multiplicity, and let $\mathcal{F}$ be any entire function of exponential type whose zero set is $\mathscr{X}_{\boldsymbol{\gamma}}$, with multiplicities. Then $|\mathcal{F}(\mathrm{x} + i\hbar)|^2 \notin (A_2)$ for every real $\hbar \neq 0$. The proof does not assume the Riemann hypothesis.*

Proof. Write $\Psi := \Psi_{\boldsymbol{\gamma}}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ for the function of Definition 6.1. It is even and real by Lemma 5.2(iii), because $\mathrm{x}_1 = \mathcal{N}_{\mathrm{RVM}}(\gamma_1) = 0.45\ldots > 0$, so $0 \notin \mathscr{X}_{\boldsymbol{\gamma}}$.

Step 1: reduction to $\mathcal{F}_{\boldsymbol{\gamma}}$ and $\hbar > 0$. $\mathcal{F}$ and $\mathcal{F}_{\boldsymbol{\gamma}}$ are entire of order at most 1 (exponential type) with the same zeros, with multiplicity, so by Hadamard's factorisation theorem both are e^{linear} times the same canonical product, and $\mathcal{F}/\mathcal{F}_{\boldsymbol{\gamma}} = e^{\mathrm{a}_1 + \mathrm{b}_1 \mathfrak{z}}$. Hence $|\mathcal{F}(\mathrm{x} + i\hbar)|^2 = \mathrm{c}\, e^{2\mathrm{b}_1' \mathrm{x}} \mathrm{w}_0(\mathrm{x})$ with $\mathrm{b}_1' = \Re \mathrm{b}_1$, a constant $\mathrm{c} > 0$, and $\mathrm{w}_0(\mathrm{x}) := |\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x} + i\hbar)|^2$, which is even in x. Moreover $|\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x} - i\hbar)| = |\mathcal{F}_{\boldsymbol{\gamma}}(\mathrm{x} + i\hbar)|$. Suppose $e^{2\mathrm{b}_1' \mathrm{x}} \mathrm{w}_0 \in (A_2)$. Reflection preserves (A_2), so $e^{-2\mathrm{b}_1' \mathrm{x}} \mathrm{w}_0 \in (A_2)$ as well. By Step 2 below both logarithms are in BMO, so $4\mathrm{b}_1' \mathrm{x} \in \mathrm{BMO}$, which forces $\mathrm{b}_1' = 0$. So we may assume $\mathcal{F} = \mathcal{F}_{\boldsymbol{\gamma}}$ and $\hbar > 0$, and that $\mathrm{w}_0 \in (A_2)$.

Step 2: BMO. Let $\mathrm{f} := \log \mathrm{w}_0$ and f_I its mean on I. By Jensen's inequality, $\frac{1}{|I|} \int_I e^{\pm(f - \mathrm{f}_I)} \geq 1$. The product of the two averages is at most $[\mathrm{w}_0]_{A_2}$, so each is at most $[\mathrm{w}_0]_{A_2}$. Since $u^2 \leq 2e^{|u|} \leq 2(e^u + e^{-u})$,

$$\frac{1}{|I|} \int_I |f - \mathrm{f}_I|^2 \leq 4[\mathrm{w}_0]_{A_2} \qquad \text{for every interval } I. \tag{6.8}$$

By (5.6) and the boundedness of $\log|\mathcal{F}_0(\mathrm{x} + i\hbar)|$, $\mathrm{g}_0 := \pi \mathbb{Y}_\hbar \Psi \in \mathrm{BMO}$. Now take $\mathbb{A}_\hbar(\mathfrak{z}) := \mathbb{A}_{\mathscr{X}}(\mathfrak{z} + i\hbar)$, in the notation of the proof of Lemma 5.2, which is holomorphic on $\Im \mathfrak{z} > -\hbar$. Its real part on $\mathbb{R}$ is g_0, and its imaginary part on $\mathbb{R}$ is $-\pi \mathbb{P}_\hbar \Psi$. By (5.7), $\Re \mathbb{A}_\hbar(\mathfrak{z}) = O(\log^2(4 + |\mathfrak{z}|))$ on $\Im \mathfrak{z} \geq 0$, and so is the Poisson integral $\mathbb{P}[\mathrm{g}_0]$. Hence $\Re \mathbb{A}_\hbar - \mathbb{P}[\mathrm{g}_0]$ is harmonic on $\Im \mathfrak{z} > 0$, continuous with zero boundary values, and $O(\log^2 |\mathfrak{z}|)$. After odd reflection it is harmonic on $\mathbb{C}$ and $o(|\mathfrak{z}|)$, hence constant, hence zero. Therefore $\mathbb{A}_\hbar = \mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0] + ic$ on $\Im \mathfrak{z} > 0$, where $\mathbb{P}[\mathrm{g}_0] + i\mathbb{Y}[\mathrm{g}_0]$ is the holomorphic function with real part $\mathbb{P}[\mathrm{g}_0]$ and $\mathbb{Y}[\mathrm{g}_0](\mathrm{x} + i\mathrm{y}) = \mathbb{Y}_\mathrm{y} \mathrm{g}_0(\mathrm{x})$. Letting $\mathrm{y} \to 0$, $-\pi \mathbb{P}_\hbar \Psi = \tilde{\mathrm{g}}_0 + c$ almost everywhere on $\mathbb{R}$, where $\tilde{\mathrm{g}}_0$ is the conjugate function of g_0, the a.e. boundary value of $\mathbb{Y}_\mathrm{y} u$ [9, Ch. III]. Conjugation is bounded on BMO [9, Ch. VI], so $\mathbb{P}_\hbar \Psi \in \mathrm{BMO}$. By the John–Nirenberg inequality [9, Ch. VI], there is $\mathrm{c}_{\mathrm{JN}} < \infty$ with

$$\frac{1}{|I|} \int_I \left| \mathbb{P}_\hbar \Psi - (\mathbb{P}_\hbar \Psi)_I \right|^2 \leq \mathrm{c}_{\mathrm{JN}} \qquad \text{for every interval } I. \tag{6.9}$$

Step 3: back to the ordinates. By [17, §9.3], $N_{\boldsymbol{\gamma}}(t) = 1 + \vartheta(t)/\pi + S(t)$, with $S(t) = O(\log t)$ [17, Thm. 9.4]. Lemma 6.2 and (2.2) then give, for $\mathrm{x} = \mathcal{N}_{\mathrm{RVM}}(t) > 0$, $\Psi(\mathrm{x}) = S(t) + \mathrm{r}(\mathrm{x})$ with $|\mathrm{r}| \leq \mathrm{c}_e$. Here c_e bounds $|1 + \vartheta/\pi - \mathcal{N}_{\mathrm{RVM}}| + \frac{1}{2}$ on $t \geq 2\pi$. Let $\mathfrak{t}_0 := \mathcal{N}_{\mathrm{RVM}}^{-1}(0)$ and $T \to \infty$, with $t \in [T, 2T]$ and $\hbar_t := \hbar/\mathcal{N}_{\mathrm{RVM}}'(t)$, $\hbar_T := \hbar/\mathcal{N}_{\mathrm{RVM}}'(T)$.

(a) The part of $\mathbb{P}_\hbar \Psi(\mathcal{N}_{\mathrm{RVM}}(t))$ coming from $\mathrm{x}' < 0$ is $O(\log T/T)$, since $|\Psi(\mathrm{x}')| \leq \mathrm{c} \log(2 + |\mathrm{x}'|)$. The part coming from r is at most c_e. Substituting $\mathrm{x}' = \mathcal{N}_{\mathrm{RVM}}(\mathrm{s})$, the rest is $\int_{\mathfrak{t}_0}^\infty \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) S(\mathrm{s})\, d\mathrm{s}$ with $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) := \mathbb{P}_\hbar(\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}))\mathcal{N}_{\mathrm{RVM}}'(\mathrm{s})$, where $\mathbb{P}_\hbar(\mathrm{v}) := \frac{\hbar}{\pi(\mathrm{v}^2 + \hbar^2)}$.

(b) Put $\mathbb{P}_t^\circ(\mathrm{s}) := \mathbb{P}_{\hbar_t}(t - \mathrm{s}) = \mathbb{P}_\hbar(\mathcal{N}_{\mathrm{RVM}}'(t)(t - \mathrm{s}))\mathcal{N}_{\mathrm{RVM}}'(t)$. For $|\mathrm{s} - t| \leq T^{1/3}$, Taylor's formula with $\mathcal{N}_{\mathrm{RVM}}'' = \frac{1}{2\pi \mathrm{s}}$ gives $|\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}_{\mathrm{RVM}}'(t)(t - \mathrm{s})| \leq \mathrm{d}_T := T^{-1/3}/\pi$. Hence $|\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) - \mathbb{P}_t^\circ(\mathrm{s})| \leq (5\mathrm{d}_T/\hbar)\mathbb{P}_t^\circ(\mathrm{s})$ for T large. With $|S| \leq \mathrm{c} \log(3T)$ on that range, the core difference is $O(T^{-1/3} \log T)$. For $|\mathrm{s} - t| > T^{1/3}$ we use $\mathbb{P}_\hbar(\mathrm{v}) \leq \frac{\hbar}{\pi \mathrm{v}^2}$, the convexity of $\mathcal{N}_{\mathrm{RVM}}$

and $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$. If $\mathrm{s} > t + T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) - \mathcal{N}_{\mathrm{RVM}}(t) \ge \mathcal{N}'_{\mathrm{RVM}}(t)(\mathrm{s}-t) \ge \mathrm{s}-t$ for T large, so $\int_{t+T^{1/3}}^{\infty} \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s})|S(\mathrm{s})|\,d\mathrm{s} \le \frac{\mathrm{c}\hbar}{\pi}\int_{T^{1/3}}^{\infty} \frac{\mathcal{N}'_{\mathrm{RVM}}(t+\mathrm{v})\log(2+t+\mathrm{v})}{\mathrm{v}^2}\,d\mathrm{v} = O(T^{-1/3}\log^2 T)$, splitting the v-integral at $\mathrm{v}=T$. If $\mathfrak{t}_0 \le \mathrm{s} < t - T^{1/3}$, then $\mathcal{N}_{\mathrm{RVM}}(t) - \mathcal{N}_{\mathrm{RVM}}(\mathrm{s}) \ge \mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})(t-\mathrm{s})$ and $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s}) \ge \mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0) > 0$, so $\mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}(\mathrm{s}) \le \frac{h}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})(t-\mathrm{s})^2} \le \frac{h}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0)(t-\mathrm{s})^2}$ and $\int_{\mathfrak{t}_0}^{t-T^{1/3}} \mathbb{P}_t^{\mathcal{N}_{\mathrm{RVM}}}|S| \le \frac{\mathrm{c}\hbar\log(3T)}{\pi\mathcal{N}'_{\mathrm{RVM}}(\mathfrak{t}_0)}\int_{T^{1/3}}^{\infty}\mathrm{v}^{-2}\,d\mathrm{v} = O(T^{-1/3}\log T)$. The same bounds hold for $\mathbb{P}_t^{\circ}$ with $\hbar_t$ in place of $\hbar$ and $\mathcal{N}'_{\mathrm{RVM}}(t)$ in place of $\mathcal{N}'_{\mathrm{RVM}}(\mathrm{s})$, the region $\mathrm{s} \gg T$ being handled by the split at $\mathrm{v}=T$. So that part tends to 0 as well.

(c) $\|\mathbb{P}_a - \mathbb{P}_b\|_{L^1} \le \frac{2}{\pi}|\log(a/b)|$, because $\int|\partial_c \mathbb{P}_c| = \frac{2}{\pi c}$. Since $\hbar_t/\hbar_T \in [1 - \frac{2\log 2}{\log(T/2\pi)}, 1]$, $|\int(\mathbb{P}_{\hbar_t} - \mathbb{P}_{\hbar_T})(t-\mathrm{s})S(\mathrm{s})\,d\mathrm{s}| = O(\log T/\log T) = O(1)$, again with the tails controlled by $|S(\mathrm{s})| \le \mathrm{c}\log(2+\mathrm{s})$.

(d) $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O(\sqrt{T})$, with $S := 0$ on $(-\infty, \mathfrak{t}_0)$; this is Lemma 6.9 below.

Together, (a)–(d) give $\|\mathbb{P}_\hbar\Psi \circ \mathcal{N}_{\mathrm{RVM}} - S\|_{L^2[T,2T]} = O(\sqrt{T})$.

Step 4: contradiction. Let $\mathfrak{I}_T := [\mathcal{N}_{\mathrm{RVM}}(T), \mathcal{N}_{\mathrm{RVM}}(2T)]$ and $\bar{\mathrm{p}}_T := (\mathbb{P}_\hbar\Psi)_{\mathfrak{I}_T}$. By (6.9), $\int_T^{2T} |\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) - \bar{\mathrm{p}}_T|^2 \mathcal{N}'_{\mathrm{RVM}}(t)\,dt \le \mathrm{c}_{\mathrm{JN}}|\mathfrak{I}_T| \le \mathrm{c}_{\mathrm{JN}} T\mathcal{N}'_{\mathrm{RVM}}(2T)$, so $\int_T^{2T}|\mathbb{P}_\hbar\Psi(\mathcal{N}_{\mathrm{RVM}}(t)) - \bar{\mathrm{p}}_T|^2\,dt \le 2\mathrm{c}_{\mathrm{JN}}T$. With Step 3, $\int_T^{2T}|S-\bar{\mathrm{p}}_T|^2 = O(T)$. But

$$\int_T^{2T}|S-\mathrm{c}|^2 \ge \int_T^{2T} S^2 - \frac{1}{T}\Big(\int_T^{2T} S\Big)^2 = \frac{1}{2\pi^2}T\log\log T\,(1+o(1)) \tag{6.10}$$

for every constant c, by Selberg's theorem $\int_0^T S^2 = \frac{T}{2\pi^2}\log\log T + O(T(\log\log T)^{1/2})$ [16], with the normalisation $S(t) = \pi^{-1}\arg\zeta(\frac{1}{2}+it)$ used here and in [4, p. 225], and $\int_T^{2T} S = O(\log T)$ [17, Thm. 9.9]. This is a contradiction. □

Lemma 6.9 (Smoothing S at scale $1/\log T$)**.** *Let $\hbar > 0$ be fixed, $\hbar_T := 2\pi\hbar/\log(T/2\pi)$, and $S := 0$ on $(-\infty, \mathfrak{t}_0)$. Then $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} = O_\hbar(\sqrt{T})$, unconditionally.*

Proof. Fujii's Main Theorem [4, Main Theorem], proved unconditionally by modifying Selberg's method, states for $q=1$, $k=1$, $T^{1/2+a} \le H \le T$ and $0 < u \le H - (H/\sqrt{T})^{1/8}$ that $\int_T^{T+H}(S(t+u)-S(t))^2dt = c_1 H\cdot 2\log(2+u\log T)\,(1+O((\log(2+u\log T))^{-1/2}))$ with an absolute c_1. The relative error is $O(1)$ when $u\log T$ is bounded, so the statement carries no asymptotic information there, and we use it only as the upper bound it contains: for $0 < u \le 1$, $T \ge T_0$ and $H = T$,

$$\int_T^{2T}\big(S(t+u)-S(t)\big)^2dt \le \mathrm{c}_F T\log(2+u\log T), \tag{6.11}$$

with an absolute constant c_F. The range $0 < u \le H - (H/\sqrt{T})^{1/8} = T - T^{1/16}$ of [4] contains $0 < u \le 1$, and the same bound holds for $-1 \le u < 0$: the substitution $t \mapsto t-u$ moves the integral to $[T-|u|, 2T-|u|] \subset [T/2, 2T]$, covered by (6.11) at the base points $T/2$ (with $H = T/2$) and T (with $H = T$). By Minkowski's inequality, $\|\mathbb{P}_{\hbar_T} * S - S\|_{L^2[T,2T]} \le \int \mathbb{P}_{\hbar_T}(u)\|S(\cdot - u) - S\|_{L^2[T,2T]}\,du$. For $0 < |u| \le 1$ the norm is at most $(2\mathrm{c}_F T\log(2+|u|\log T))^{1/2}$. For $1 < |u| \le T/2$, Selberg's theorem $\int_0^T S^2 = \frac{1}{2\pi^2}T\log\log T + O(T(\log\log T)^{1/2})$ [16] bounds it by $\mathrm{c}(T\log\log T)^{1/2}$. For $|u| > T/2$, $|S(t)| \le \mathrm{c}\log(2+t)$ [17, Thm. 9.4] bounds it by $\mathrm{c}\sqrt{T}\log(3T+|u|)$. Substituting $u = \hbar_T v$ in the first range, where $\hbar_T\log T \to 2\pi\hbar$, and using $\int_{|u|>1}\mathbb{P}_{\hbar_T} = O(\hbar_T)$ and $\int_{|u|>T/2}\mathbb{P}_{\hbar_T}(u)\log(3T+|u|)\,du = O(\hbar_T\log T/T)$, all three contributions are $O(\sqrt{T})$. □

Remark 6.10 (A second proof). Lemma 6.9 also follows from Selberg's mean-square approximation of $S(t)$ by $S_{\mathrm{x}_0}(t) := -\frac{1}{\pi}\sum_{p\le \mathrm{x}} p^{-1/2}\sin(t\log p)$, $\mathrm{x}_0 = T^\delta$ [16], and the Montgomery–Vaughan mean value theorem [14, Cor. 3]. Indeed $\mathbb{P}_{\hbar_T} * S_{\mathrm{x}_0} - S_{\mathrm{x}_0} = -\frac{1}{\pi}\sum_{p\le \mathrm{x}} p^{-1/2}(e^{-\hbar_T\log p} - 1)\sin(t\log p)$ has mean square $\ll T\,\hbar_T^2\log^2\mathrm{x} \ll T$ on $[T,2T]$, and the remainder $S - S_{\mathrm{x}_0}$ is handled by Minkowski's inequality. Fujii's announcement [5] states the moment lemma of [4] in the form used in [6, Lemma 1].

Corollary 6.11 (Consequences for the zeta list)**.** (a) $\{e^{ix_k\xi}\}_{x_k\in\mathscr{X}_\gamma}$ *is not a Riesz basis of* $L^2(-\pi,\pi)$. (b) *If* $\mathscr{X}_\gamma$ *is a finite union of uniformly discrete sets, its generalised divided differences are not a Riesz basis of* $L^2(-\pi,\pi)$.
(c) *For every decomposition* $\Psi_\gamma = A + B$ *with* $\|A\|_\infty < \frac14$, $\mathbb{Y}_1 B$ *is unbounded. In particular this holds for the blocks of Proposition 6.3, where the averaging hypothesis is satisfied.*

Proof. (a) and (b): the criteria [3, Thm. 1, Thm. 3] are necessary and sufficient, and require a generating function satisfying Helson–Szegő, equivalently (A_2), on some line $\Im\mathfrak{z} = \hbar$. The line $\hbar = 0$ is excluded, since $|\mathcal{F}(\mathrm{x})|^{-2}$ is not locally integrable at a real zero. All lines with $\hbar \neq 0$ are excluded by Theorem 6.8. (c): otherwise Theorem 5.3 would give $|\mathcal{F}_\gamma(\mathrm{x}+i)|^2 \in (A_2)$. □

Heuristic 6.12 (Graduated version)**.** The proof of Theorem 6.8 uses intervals $\mathfrak{I}_T$ of length $\asymp T\log T$. On a window of L unfolded points at height T, the mean square oscillation of $\mathbb{P}_\hbar\Psi$ is governed by the variance of increments of S over t-intervals of length $L/\mathcal{N}'_{\mathrm{RVM}}(T)$. By Fujii's asymptotics this is expected to be of size $\frac{1}{\pi^2}\log L$. This suggests that local (A_2) constants on windows of L points grow with L but stay bounded for L bounded. That is the graduated stability that a Fredholm formulation would need. None of this is proved here. The computations of Section 7 measure the ℓ^∞ norm of the inverse of square finite sections directly and find growth like $T^{1.8}$ up to $T = 487$.

Remark 6.13 (Bearing on route B)*.* Theorem 6.8 rules out a uniform Riesz-basis (Fredholm) structure for the unfolded zeta list on the whole line, and hence a linear stability argument with constants independent of the height. It says nothing about statement B or B′. The compactness step of route B (Proposition 3.4) uses no stability; the open part of route B is existence (Definition 3.1) together with uniqueness in the box (Hypothesis 3.5), and the open part of route B′ is existence alone (Corollary 3.14).

7. Numerical certificates for finite sections

This section reports computations. They are not part of any proof; their role is to measure the quantities of Theorems 3.19 and 3.20 on actual sections and to locate where the staged construction fails. The programs and their outputs are listed in the appendix of the main manuscript and in the archive accompanying this note. Floating point arithmetic was used throughout; no interval arithmetic was run.

7.1. **Set-up.** The tests are Gaussians $\varphi_{c,\mathrm{w}}(u) := e^{-(u-c)^2/2\mathrm{w}^2}$ with centre $c \geq 9\mathrm{w}$, so that their restriction to $u > 0$ is, to working precision, a test supported in $(0,\infty)$; their cosine transform is $\mathsf{c}_{\varphi_{c,\mathrm{w}}}(t) = \mathrm{w}\sqrt{2\pi}\,e^{-(\mathrm{w}t)^2/2}\cos(ct)$. Stage m uses the width $\mathrm{w}_m := 0.08\cdot 0.85^{m-1}$, the height $T_m := 4/\mathrm{w}_m$, and the new tests φ_{c,w_m} with centres $c \in [9\mathrm{w}_m,\, \log(\max(T_m,7)/2\pi)+2]$ in steps of w_m; the test set $\mathscr{T}_m$ is the union of the levels $\leq m$. The unknowns of stage m are the ordinates with $\tau_k \leq 2.4/\mathrm{w}_m$ ("interior"); the factor $e^{-(\mathrm{w}_m\kappa)^2/2}$ makes the others invisible to the tests of level m. The masses are a_n^ζ for all $n \leq e^{c+9\mathrm{w}}$, and k is `eq:transport-defs`. The pairings $\langle \mathsf{k}, \varphi\rangle$ were computed by adaptive quadrature. The zeta ordinates γ_k, $k \leq 400$ ($\gamma_{400} = 679.74$), computed with `mpmath`, enter only in the diagnostic columns; no construction uses them.

Two checks fix the normalisation: $\langle \mathfrak{R}(\boldsymbol{\gamma}, \mathsf{a}^\zeta), \varphi\rangle \approx 10^{-16}$ for all tests tried, and $\langle \mathfrak{R}(\boldsymbol{\tau}, \mathsf{a}^\zeta), \varphi\rangle$ between 10^{-3} and 10^{-1}.

7.2. **Initialisation: the clock is outside the basin, the Gonek lists are inside.** *Receding horizon from the clock.* Starting at $\boldsymbol{\kappa} = \boldsymbol{\tau}$, each stage solved the overdetermined least-squares problem for the interior ordinates with all tests of levels $\leq m$, with bounds $|\kappa_k - \tau_k| \leq 3$ and the non-interior ordinates frozen at their current values. The iteration stalls at the first Gram-type failure of the clock: the fit returns $\kappa_{34} = 109.75$ where $\gamma_{34} = 111.03$ ($\tau_{34} = 110.04$), and

$\kappa_{41} = 125.50$ where $\gamma_{41} = 124.26$; from $m = 9$ on the residual on the lagged levels $\le m - 4$ grows from $2 \cdot 10^{-5}$ to $4 \cdot 10^{-2}$ and the front of ordinates within 10^{-2} of $\boldsymbol{\gamma}$ stays at $k \approx 31$.

Receding horizon from a Gonek list. The same iteration started at $\boldsymbol{\kappa}^{(\mathrm{X})}$ with $\mathrm{X} = 8$ (Lemma 3.25; 983 roots up to $t = 1400$, computed from the Riemann–Siegel expansion of ϑ and the sum (3.15)) converged through $m = 16$, $T_{16} = 572$, 165 interior ordinates and 4985 tests: all interior ordinates within 10^{-2} of γ_k, 90 of them within 10^{-4}, the first twenty within $5 \cdot 10^{-6}$, residual on the lagged levels between 10^{-6} and $9 \cdot 10^{-6}$ at every stage, and $\max_k |\kappa_k - \tau_k| = 0.99$ throughout. The residual over all tests of level $\le m$ stays at $2 \cdot 10^{-3}$, which is the contribution of the frozen edge.

Why. The basin of the exact solution, measured by random perturbations of $\boldsymbol{\gamma}$ in the interior block, has radius 0.1 to 0.25 mean spacings. The clock lies at up to 0.66 spacings from $\boldsymbol{\gamma}$ for $k \le 400$, with 50 indices beyond 0.25; the lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ lie within 0.184 ($\mathrm{X} = 3$), 0.081 ($\mathrm{X} = 5$), 0.053 ($\mathrm{X} = 8$), 0.047 ($\mathrm{X} = 12$) and 0.035 ($\mathrm{X} = 20$) spacings for $k \le 160$, and within 0.103 spacings for $k \le 400$ at $\mathrm{X} = 8$ (Figure 1). The ℓ^2 conditioning of the interior block at $\boldsymbol{\gamma}$ is moderate and roughly uniform up to $T = 299$: condition numbers 17 to 75, smallest singular value 0.08 to 0.15. The coupling of the interior to the frozen edge, $\|\mathbb{J}_I^{+}\mathbb{J}_E\|_{\infty\to\infty}$, is between 0.25 and 0.47, and the shift it induces on the low ordinates decays from $2 \cdot 10^{-2}$ at $m = 2$ to 10^{-5} for $m \ge 10$.

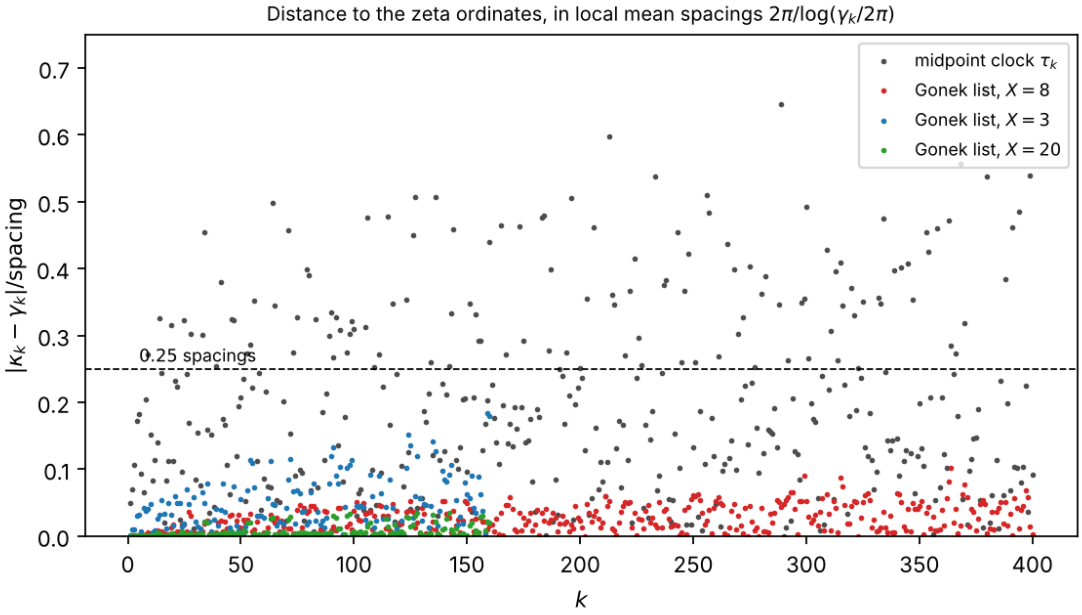


FIGURE 1. Distance to the zeta ordinates, in units of the local mean spacing $2\pi/\log(\gamma_k/2\pi)$, of the midpoint clock and of the Gonek lists $\boldsymbol{\kappa}^{(\mathrm{X})}$ for $\mathrm{X} = 3, 8, 20$ ($k \le 160$ for $\mathrm{X} = 3, 20$; $k \le 400$ for $\mathrm{X} = 8$). The dashed line is the observed radius of the Newton basin. The ordinates γ_k are used here only to measure the distance.

7.3. **Certificates at the Gonek list: the direct bounds fail.** At $\boldsymbol{\kappa}^{(8)}$, with the interior unknowns of stage m and three choices of the right inverse, the quantities of Theorem 3.19 were computed for $m = 1, \dots, 14$ (T from 50 to 414).

Right inverse and norm	Certificate index	Range over $m = 1, \dots, 14$
square subsystem (tests chosen by pivoted QR), ℓ^∞	$2\mathbb{L}_m^2\mathbb{H}_m\mathsf{h}_m$	$56 \to 1.2 \cdot 10^7$

pseudo-inverse of the full system, ℓ^2, $\beta_2 = 1/\sigma_{\min}$	$\beta_2 \mathrm{K}_2 \delta_2$ (Kantorovich)	$0.47 \to 233$
pseudo-inverse of the full system, ℓ^∞, diagonal Hessian	$2\mathrm{c}_{\mathrm{diag}}\mathsf{h}$	$1.3 \to 1.6 \cdot 10^3$

None of the certificates is satisfied, not even at $T = 50$. Yet one Gauss–Newton step from $\boldsymbol{\kappa}^{(8)}$ reduces $\max_k |\kappa_k - \gamma_k|$ by a factor between 4 and 15 at every height ($0.088 \to 0.016$ at $T = 572$), and $\sigma_{\min}$ in ℓ^2 stays between $1/17$ and $1/5$. The worst-case bounds lose a factor 10^3 to 10^4 through cancellation: $\mathbb{H}_m$ counts every unknown with the full weight $\int |\varphi| u^2$, while the actual second-order term of $\mathbb{F}_m$ along the Newton step is a sum of k_m^* oscillating terms.

7.4. **Certified exact finite sections.** The iterate of Section 7.2 was polished, stage by stage, by Newton's method on the square system (n tests selected by pivoted QR on $\mathbb{J}^T$, n interior unknowns) to residual 10^{-13}. At the polished point x_0 the certificate of Theorem 3.19 was evaluated in the sharper form that uses the separable structure of F_m: with Q the inverse of the square Jacobian, B the matrix of second derivatives $\partial^2_{\kappa_k} \mathsf{c}_\varphi$ at x_0, and $A_3(r)$ an entrywise bound for the third derivatives on $|x - x_0| \le \mathrm{r}$, the map $\mathbb{T}(x) = x - \mathbb{Q}\mathbb{F}(x)$ contracts the ball of radius r in ℓ^∞ if

$$\mathsf{l}(\mathrm{r}) := \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty}\,\mathrm{r} + \tfrac{1}{2}\,\big\||\mathbb{Q}|\mathbb{A}^{(3)}(\mathrm{r})\big\|_{\infty\to\infty}\,\mathrm{r}^2 < 1, \qquad \mathsf{h} \le (1 - \mathsf{l}(\mathrm{r}))\,\mathrm{r}, \tag{7.1}$$

in which case the ball contains an exact solution of the square system. The margin of a stage is $\max_{\mathrm{r}}(1 - \mathsf{l}(\mathrm{r}))\mathrm{r}$, the largest residual that (7.1) could certify at that point.

m	T_m	n	$\mathbb{L}_m = \|Q\|_{\infty\to\infty}$	margin	s_m on all tests $\le m$	$\max_k \|\kappa_k - \gamma_k\|$
1	50	4	41	$7.0 \cdot 10^{-3}$	$7.7 \cdot 10^{-8}$	$7 \cdot 10^{-7}$
4	81	9	50	$3.1 \cdot 10^{-3}$	$2.6 \cdot 10^{-7}$	$1 \cdot 10^{-6}$
8	156	26	150	$7.5 \cdot 10^{-4}$	$9.6 \cdot 10^{-7}$	$1 \cdot 10^{-5}$
12	299	68	460	$2.2 \cdot 10^{-4}$	$5.5 \cdot 10^{-6}$	$4 \cdot 10^{-5}$
15	487	133	1600	$6.7 \cdot 10^{-5}$	$3.3 \cdot 10^{-3}$	$4 \cdot 10^{-2}$
16	572	165	–	square Jacobian singular	–	–

Stages 1 to 15 are certified: for each, (7.1) holds with r equal to the polishing residual (10^{-15} to $5 \cdot 10^{-13}$), so an exact solution of the square section exists within r of x_0, inside the tube with $C = 1$. The margins exceed the residuals by 10^7 or more, which is far above the expected rounding error, so the same certificates are expected, but not verified, to survive interval arithmetic. The certified solutions are not unique: at $m = 15$ the solution is $4 \cdot 10^{-2}$ from $\boldsymbol{\gamma}$. At $m = 16$ the square Jacobian is singular to working precision and the stage is not certified. The scaling with height (Figure 2) is

$$\mathbb{L}_m \sim T^{1.8}, \qquad \|\mathbb{Q}\mathbb{J}'\|_{\infty\to\infty} \sim T^{2.1}, \qquad \text{margin} \sim T^{-2.1} \qquad (T \le 487). \tag{7.2}$$

7.5. **The entry step.** The open point of the staged construction, in the form that Theorem 3.20 makes precise, is whether the residual h_{m+1} of the new tests at the start of stage $m+1$ is below the margin of that stage. This was tested as follows: start from the certified solution of stage $m-1$, put the new band of unknowns (those with $\tau_k \in (2.4/\mathrm{w}_{m-1}, 2.4/\mathrm{w}_m]$) and everything beyond at their Gonek values, and evaluate (7.1) (i) for the band alone, with the new tests of level m, and (ii) for the full square system of stage m, without polishing.

The band-only test is certified for $T \le 96$ and fails from $T = 113$ on; its margin falls from $9 \cdot 10^{-2}$ to $3 \cdot 10^{-3}$ while the Gonek residual of the band stays between 10^{-2} and $3 \cdot 10^{-1}$ (the Gonek error in the band is 0.02 to 0.09). The full-system test is certified only at $T = 59$. The margin decreases as a power of T by (7.2), whereas the error of the Gonek lists with fixed X does not decrease with T, and with $\mathrm{X} = T^{\mathrm{x}}$ it would decrease at most like $1/\log T$. Within the range computed, no worst-case

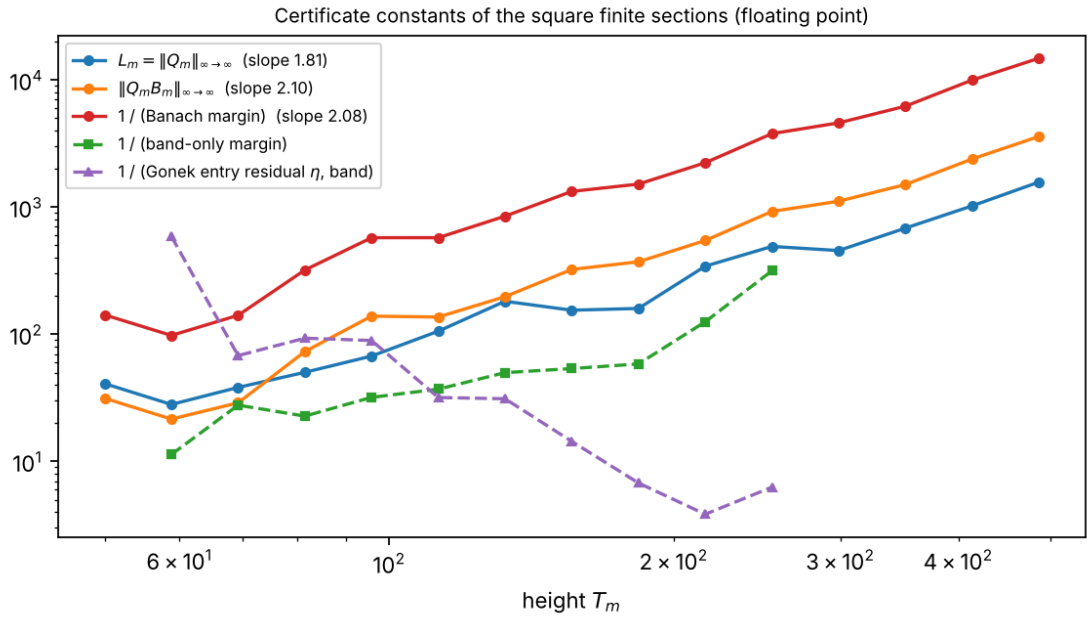


FIGURE 2. Constants of the Banach certificate (7.1) for the square sections of Section 7.4 (solid lines, with fitted slopes in $\log T$), and the band-only entry test of Section 7.5 (dashed): the reciprocal of the band margin and the reciprocal of the residual of the new band at its Gonek values. The entry test is certified where the triangles lie below the squares.

certificate in ℓ^∞ closes the entry step, and the trend of the constants says that none will at larger T. This is a numerical statement, not a theorem.

7.6. **Reading of the computations.**

(i) The constructions that use only the clock and the primes (the Gonek lists, the square sections, the certificates) behave as predicted by the tube lemma and the certificate theorem, and the certified sections exist up to $T = 487$. These sections would exist equally if the Riemann hypothesis were false: a zero $\frac{1}{2} + \mathrm{b}_0 + i\gamma_0$ off the line contributes $2\cosh(\mathrm{b}_0 u)\cos(\gamma_0 u)$ to the prime side, which real ordinates near γ_0 can imitate against tests supported in $(0, u^*)$ as long as $\mathrm{b}_0 u^*$ is small. The information about the hypothesis sits entirely at heights $T \gtrsim (C \log \gamma_0)^{1/\mathrm{b}_0}$, which no finite computation reaches.

(ii) The quantities $\mathbb{L}_m$ and $\|QB\|$ grow polynomially in T. This is the quantitative face of the obstruction theorem of Section 6.3: no height-uniform linear stability, hence no height-uniform Newton basin. A certificate for the infinite construction needs a basin that shrinks no faster than the initial error of a γ-free list decreases, and the computed trend is T^{-2} against $1/\log T$.

(iii) The observed contraction of Gauss–Newton (factor 4 to 15 per step at every height) is a mean-square phenomenon that the worst-case constants do not see. A proof of the entry step would have to bound $\|\mathbb{Q}\mathbb{J}' \operatorname{diag}(\mathrm{e})\, \mathrm{e}\|_\infty$ for the actual error vector e of the Gonek list, not for an arbitrary vector of the same norm; that error vector is γ-free, and its residual $\mathbb{F}_m(\boldsymbol{\kappa}^{(\mathrm{X})})$ is controlled through the explicit formula by the moments of $S - S_{\mathrm{X}}$, which are unconditional (Selberg, Fujii). Whether this closes is not known; any proof along this line would contain a proof of the Riemann hypothesis, by Corollary 3.14.

8. Status of the statements

Statement	Status
Lemma 3.2, Corollary 3.3	proved; conditional on (D) as stated
Proposition 3.4, Lemma 3.17	proved, unconditional
Hypothesis 3.5	open; it extends `thm:free-masses`; not needed with prescribed masses
Corollary 3.7	proved, assuming B and U
Lemmas 3.9, 3.10	proved, unconditional (the former Hypothesis (L))
Proposition 3.12, Corollary 3.14	proved, unconditional: RH $\Leftrightarrow$ B$'(C)$; classical in substance
Lemma 3.18, Theorems 3.19, 3.20, Proposition 3.22	proved; their hypotheses (A$_m$) for all m and (B) are open and imply RH
Heuristic 3.24	not proved; its point model is not a test of $C_c^\infty(0,\infty)$
Lemma 3.25	proved, unconditional; uses [7] only for the interpretation of the list
(G1), (G2) of Remark 3.26	open; together they imply B$'$ and RH
Lemmas 4.2–4.7, Theorem 4.8, Corollary 4.9	proved, unconditional; uses `thm:halfplane-RVM`
Heuristics 4.11, 4.12, 6.12	not proved
Lemma 5.2, Theorem 5.3, Corollary 5.4	proved; Corollary 5.4 uses [3]
Corollary 5.6, Theorem 5.7, Proposition 5.5	proved; novelty not checked
Lemma 6.2, Proposition 6.3, Proposition 6.6, Corollary 6.5	proved, unconditional
Lemma 6.9, Theorem 6.8, Corollary 6.11	proved, unconditional; uses the upper bound contained in [4, Main Theorem], [16, 17, 9]
Section 7	computations in floating point; not part of any proof

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